XQ-37 Reaver

Homeland Defense Interceptor Design Proposal

2024-25 AIAA Undergraduate Design Competition

COLLEGE OF ENGINEERING
KEVIN T. CROFTON DEPARTMENT OF
AEROSPACE AND OCEAN ENGINEERING
VIRGINIA TECH

Cyberdyne Aerospace Team

David Halston
Team Lead

Sean Rich
Chief Engineer

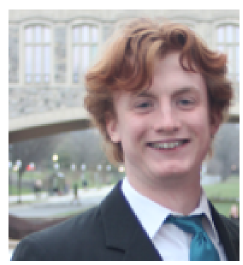
Chad Noyes
Aerodynamics Lead

Laurel Hudson
Structures Lead

Andrew Adams
Propulsion Lead

Jeffrey Roback
Utilities & Subsystems Lead

Garrett Mongold
System Integration Lead



Eli Campbell
Mission Systems & Evaluation Lead

Contents

List of Tables

List of Figures

Nomenclature

A_c	=	cross-sectional area of the inlet, capture area
A_∞	=	cross-sectional area of freestream air entering the inlet
A_{∞_E}	=	engine demand cross-sectional area
$\dot{m}_c$	=	inlet capture area airflow
$\dot{m}_E$	=	engine airflow
$\dot{m}_S$	=	secondary airflow
$\dot{m}_\infty$	=	freestream airflow
V_∞	=	freestream velocity
ρ_∞	=	freestream density
$\bar{V}_H$	=	horizontal tail volume coefficient
l	=	distance from wing aerodynamic center to horizontal tail aerodynamic center
L	=	aircraft overall length
$\bar{V}_V$	=	vertical tail volume coefficient
$C_{D_{\text{wave}}}$	=	wave drag coefficient
A_{max}	=	maximum frontal area
E_{WD}	=	wave drag efficiency factor
Λ_{LE}	=	leading edge sweep
M	=	Mach number
$M_{C_{D_0\,\text{max}}}$	=	Mach of maximum zero-lift drag
S_{TO}	=	takeoff distance
S_L	=	landing distance
S_{FR}	=	landing freeroll distance
S_B	=	braking distance
S_A	=	airborne distance
σ	=	takeoff parameter
$C_{L_{\text{max}}}$	=	maximum coefficient of lift
θ_{app}	=	approach angle
μ	=	coefficient of friction
V_{TD}	=	touchdown speed
V_{TO}	=	takeoff speed
V_{50}	=	velocity over 50 ft obstacle
C_{D0}	=	zero-lift drag coefficient
C_{LG}	=	ground roll lift
$\Delta C_{D_{\text{flaps}}}$	=	drag increase due to flaps
$\Delta C_{D_{\text{gear}}}$	=	drag increase due to gear
$\Delta C_{D_{\text{misc}}}$	=	drag increase due to misc
$\Delta C_{D_{\text{spoilers}}}$	=	drag increase due to spoilers
ψ	=	overturn angle
h_{cg}	=	cg height
l_n	=	distance from nose landing gear to cg
$C_{M\alpha}$	=	static longitudinal (pitch) stability derivative
$C_{l\beta}$	=	static lateral (roll) stability derivative

$C_{n\beta}$	=	static directional (yaw) stability derivative
ζ_{ph}	=	phugoid mode damping ratio
ζ_s	=	short period mode damping ratio
T_R	=	roll subsidence mode
ζ_{dr}	=	dutch roll mode damping ratio
T_{2s}	=	spiral mode

Acronyms

$AESA$	=	Active Electronically Scanned Array
$AIAA$	=	American Institute of Aeronautics and Astronautics
AIU	=	Antenna Interface Unit
AC	=	Aerodynamic Center
AI	=	Artificial Intelligence
$BLOS$	=	Beyond Line-of-Sight
CAD	=	Computer Aided Design
CAP	=	Counter-Air Patrol
$C2S$	=	Command and Control Station
CG	=	Center of Gravity
DAS	=	Digital Aperture System
DOD	=	Department of Defense
DoA	=	Description of Action
DSI	=	Diverterless Supersonic Intake
DPO	=	Data Protection Officer
EAS	=	Equivalent Airspeed
EC	=	European Commission
EHA	=	Electro-Hydrostatic Actuator
$EOTS$	=	Electro-Optical Targeting System
ESR	=	Early Stage Researcher
$FAIR$	=	Findable, Accessible, Interoperable, Reusable
GCS	=	Ground Control Station
$GDPR$	=	General Data Protection Regulation
ICP	=	Integrated Core Processor
ISR	=	Intelligence, Surveillance, and Reconnaissance
$JADC2$	=	Joint All Domain Command and Control
$JADO$	=	Joint All-Domain Operations
KPI	=	Key Performance Indicator
LEO	=	Low Earth Orbit
LOS	=	Line-of-Sight
$MADL$	=	Multifunctional Advanced Data Link
MSL	=	Mean Sea Level
$MIDS$	=	Multifunctional Information Distribution System
$PTMS$	=	Power and Thermal Management System
RFP	=	Request for Proposal
RRI	=	Responsible Research and Innovation
$SATCOM$	=	Satellite Communications
SLS	=	Sea Level Static
SM	=	Static Margin
$SRIA$	=	Strategic Research and Innovation Agenda
SSH	=	Social Sciences and Humanities
$TCDL$	=	Tactical Common Data Link

TBD	=	To Be Determined
$TOGW$	=	Take-Off Gross Weight
$TSFC$	=	Thrust Specific Fuel Consumption
$USAF$	=	United States Air Force
VMC	=	Vehicle Management Computer
VMS	=	Vehicle Management System

1 Executive Summary

As international tensions continue to rise, an increasing threat of aerial attacks within the United States has contributed to a significant concern in regard to national security. Whether it be cruise missiles, foreign military drones, surveillance balloons, or even hijacked airliners, the evolving landscape of airborne threats underscores the urgent need to strengthen domestic defense capabilities. While current fifth-generation fighters are capable of countering these threats, they are too expensive to deploy on a routine basis for domestic security. On top of this, many legacy aircraft are scheduled to retire by 2045, leaving a critical gap in national aerial defense coverage.

In response to the 2025 AIAA request for proposal, Cyberdyne Aerospace is proud to present the XQ-37 Reaver. This high performance, cost effective, and remotely piloted homeland defense interceptor is purpose-built to safeguard the United States against modern aerial threats. Designed for rapid deployment with high payload capacity and advanced avionics, the XQ-37 is optimized for a broad spectrum of homeland defense missions, including defensive counter-air patrol, point defense interception, and escort operations within the national domain.

When a potential threat is identified, the XQ-37 answers the call. Able to takeoff from any 8000ft NATO-standard runway, the XQ-37 climbs from sea level to 35,000 feet in under a minute before dashing to Mach 1.6 powered by its high-performance F110-GE-129 high-bypass turbofan engine. To complement its high performance, the XQ-37 carries a robust avionics suite, featuring an active electronically scanned array radar, redundant low-latency, high bandwidth communication systems, and seamless data-link integration with manned and unmanned assets. This enables the XQ-37 to quickly find, fix, and track its target as it waits for clearance to engage its threats.

Inherently nimble, the XQ-37 can sustain between +7g and –3g turns with an instantaneous turn rate of 18 degrees per second, creating a formidable presence even against highly maneuverable, low-level threats. This agility paired with its stealth-optimized geometry gives the XQ-37 the upper hand in contested airspace. In situations that pose less of an imminent national security risk, the XQ-37 transforms into a patrolling ISR platform. With up to 4 hours of loiter time (extendable with in-air refueling) and a next-generation optical/imaging suite, the XQ-37 excels in surveillance and reconnaissance roles. Regardless of the mission, when it comes time to neutralize the target, the XQ-37 can deploy up to four AIM-120 AAMRAMs from its internal weapons bay for long range targeting, and/or two AIM-9X sidewinders for short range combat. Designed for modularity, the XQ-37's spacious internal weapons bay accommodates a broad range of ordnance, enabling mission-specific loadouts with minimal turnaround.

Post-mission, the vehicle returns with 30 minutes of fuel reserves and lands comfortably, even on icy runways, thanks to its deployable drag chute. Built for utilization across both the continental United States and NATO theaters in all-weather scenarios, the XQ-37 maximizes strategic flexibility. Most critically, the XQ-37 completes these missions unmanned. Pilots remain safe, controlling the mission from a command center, not from the cockpit. Whether the mission ends in recovery or sacrifice, no American life is ever at risk. As such, the XQ-37 is not your typical military aircraft, it represents the fearless future of autonomous homeland defense.

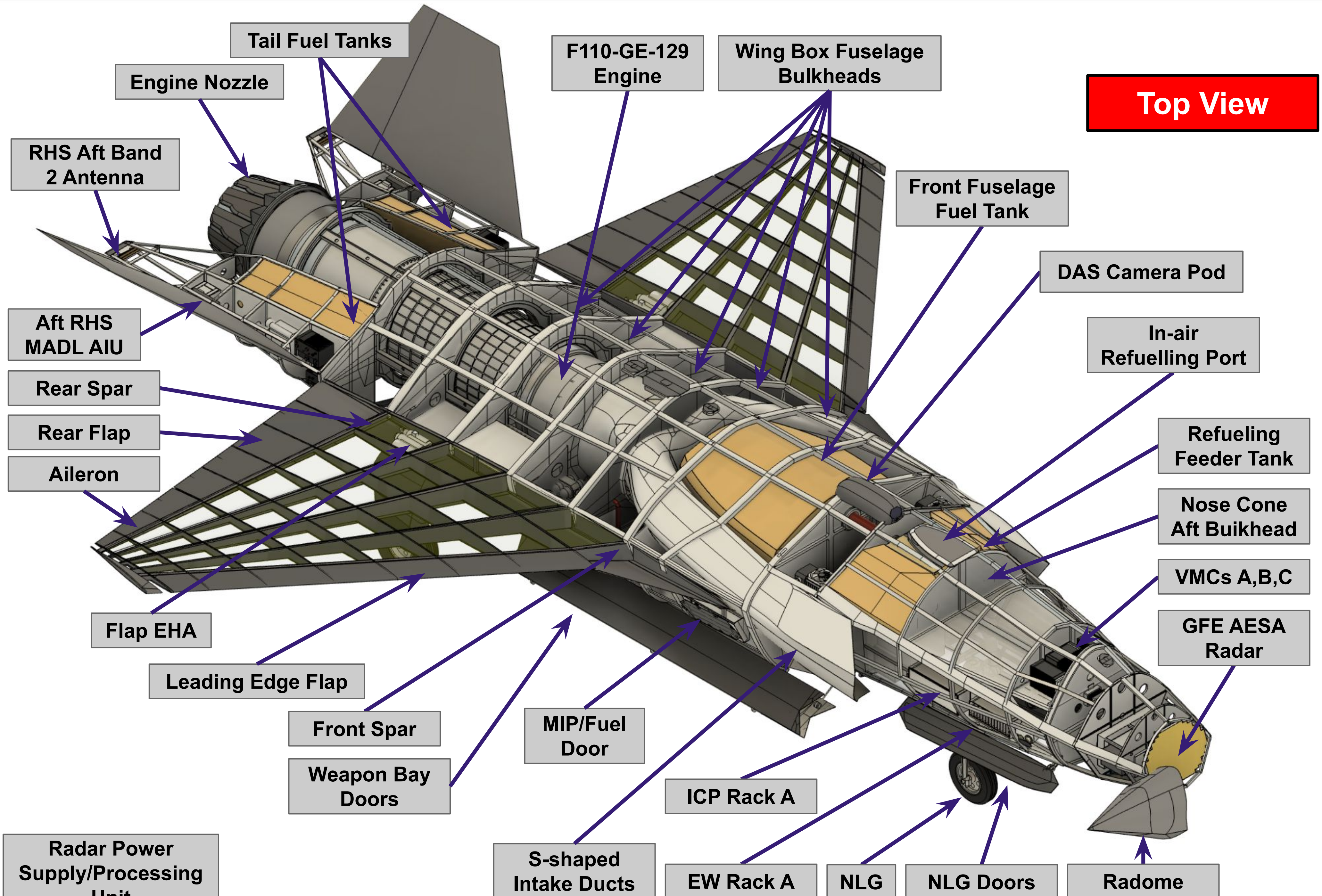
The methodology in the design of this aircraft was meticulously thoughtful and calculated. The Request for Proposal presented a wide range of explicit and implied requirements which were categorized and refined by our team into a cohesive framework highlighting the most critical aspects of our aircraft's capabilities. With these requirements, our team developed a concept that met these needs by conducting trade studies to evaluate competing priorities in order to optimize our aircraft in accordance with the RFP's core values. Throughout this process, certain requirements stood out, not only for their stringency, but also for their impact on mission success. For these distinct requirements, our team has provided a detailed breakdown demonstrating how each challenge was met, along with the rationale and evolution that brought us to our solution, ensuring the customer shares the same confidence and praise for our product that our team has had in its creation.

One category of these critical requirements revolves around robust communication systems. In particular, our team sought mission-critical redundancy with low latency, high bandwidth, and LOS, BLOS, and air-to-air communication with a particular focus on future upgradability. Given the unmanned nature of this vehicle, maintaining secure, uninterrupted communication to this vehicle is imperative. For perspective, a loss of communication without a backup system in place would result in a loss of aircraft at best, and a public safety hazard at worst. Our team took this risk seriously, developing a dependable, yet fully redundant communications architecture early in the design process to ensure seamless integration into our cost and weight analyses, maintaining budget and system performance requirements.

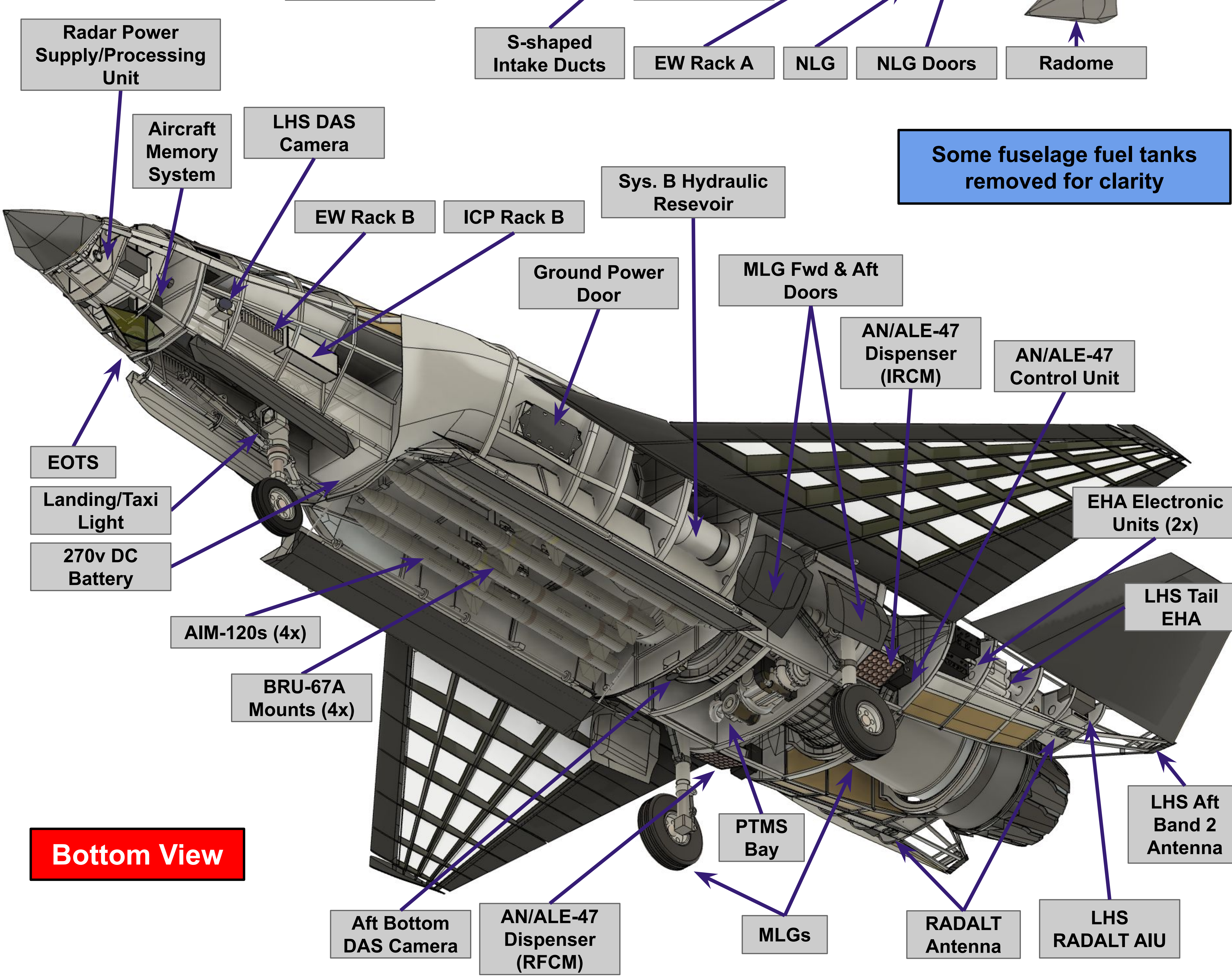
While it may seem self-evident, another critical requirement in Cyberdynes solution was that it had to complete all missions. One of the missions required by the RFP proved to be particularly challenging: the Defensive Counter-Air Patrol mission, with a segment demanding a four hour loiter. Balancing endurance, performance, and affordability introduced complex trade-offs that infiltrated multiple design domains, including fuel capacity, structures, aerodynamics, weight, balance, and stability. However, through meticulous design optimization and advanced analytical methods, the XQ-37 operates with full mission compliance, and in some cases performs with additional operational margin. This transformative design process shaped the XQ-37 into a robust platform that delivers high end capability across a broad spectrum of operational scenarios, at a fraction of the cost of next generation fighter aircraft.

The XQ-37 features a highly detailed CAD design that incorporates a full avionic suite, as well as an internal weapon bay, flight control system, fuel distribution system, sensors, hydraulics system, etc. The components were strategically placed and optimized for weight distribution, with all components' mass properties tracked and documented within the CAD program. Overall, there are over 200 parts within the CAD model, throughout over 31 major components. Some major components can be seen in the walk-around chart below.

WALKAROUND CHART



Some fuselage fuel tanks removed for clarity



2 Proposal

2.1 Understanding the Problem

This Request for Proposal highly emphasizes providing a low-cost solution, while still hitting the stringent performance requirements for proper operation of a supersonic interceptor. In order to optimize this performance without exorbitant costs, proper understanding of where this aircraft will be deployed and its context is required.

The purpose of this aircraft is to be able to cheaply and robustly bolster the defenses of the American homeland from airborne threats. In the case of a future potential conflict, modern technological advances have increased the likelihood and potential severity of kinetic and cyber-warfare attacks. Designing for the use case of defending the homeland allows for our solution to take advantage of the advanced infrastructure already established, increasing our power projection. However, in the case of overseas conflict, it is expected that many of our supporting airborne assets will be used for offensive projection. Therefore, it is critical that our solution is self-sufficient enough to be able to complete all design missions with the minimum amount of support infrastructure.

Operation of this aircraft will be centered around coastal operations, primarily the Western coast of the United States due to its proximity to near-peer nations. Here our aircraft will be able to take off from any US air force base with NATO standard runways of 8000 feet. At these airbases, proper routine maintenance on the aircraft will take place, including storage and proper handling of its munitions. Supporting the direct operation of the aircraft will include a Ground Control Station (GCS) and a Command and Control Station (C2S). The GCS will house drone pilot operators themselves with the appropriate infrastructure and support equipment to pilot the XQ-37. Meanwhile, the C2S will constantly monitor and coordinate battlefield assets in the area of operation. Constant operation will be achieved through rolling shifts. These assets together make up the direct control hub of the XQ-37.

To allow for constant battlefield updates and situational awareness, the control hub will receive updates from supporting intelligence infrastructure, mainly in the form of ground radar arrays and satellite information. The ground radar will be made up of short range mobile arrays which can be moved and pointed depending on the situation. High power long range arrays such as the NORAD array will provide additional security for tracking targets across neighboring oceans. The satellite arrays such as those operated by the US Space Force and the NRO will provide further constant operational awareness. Onboard instrumentation will provide early warning for various enemy operations, along with satellite imagery providing updates on any enemy infrastructure build-ups necessary for larger scale operations.

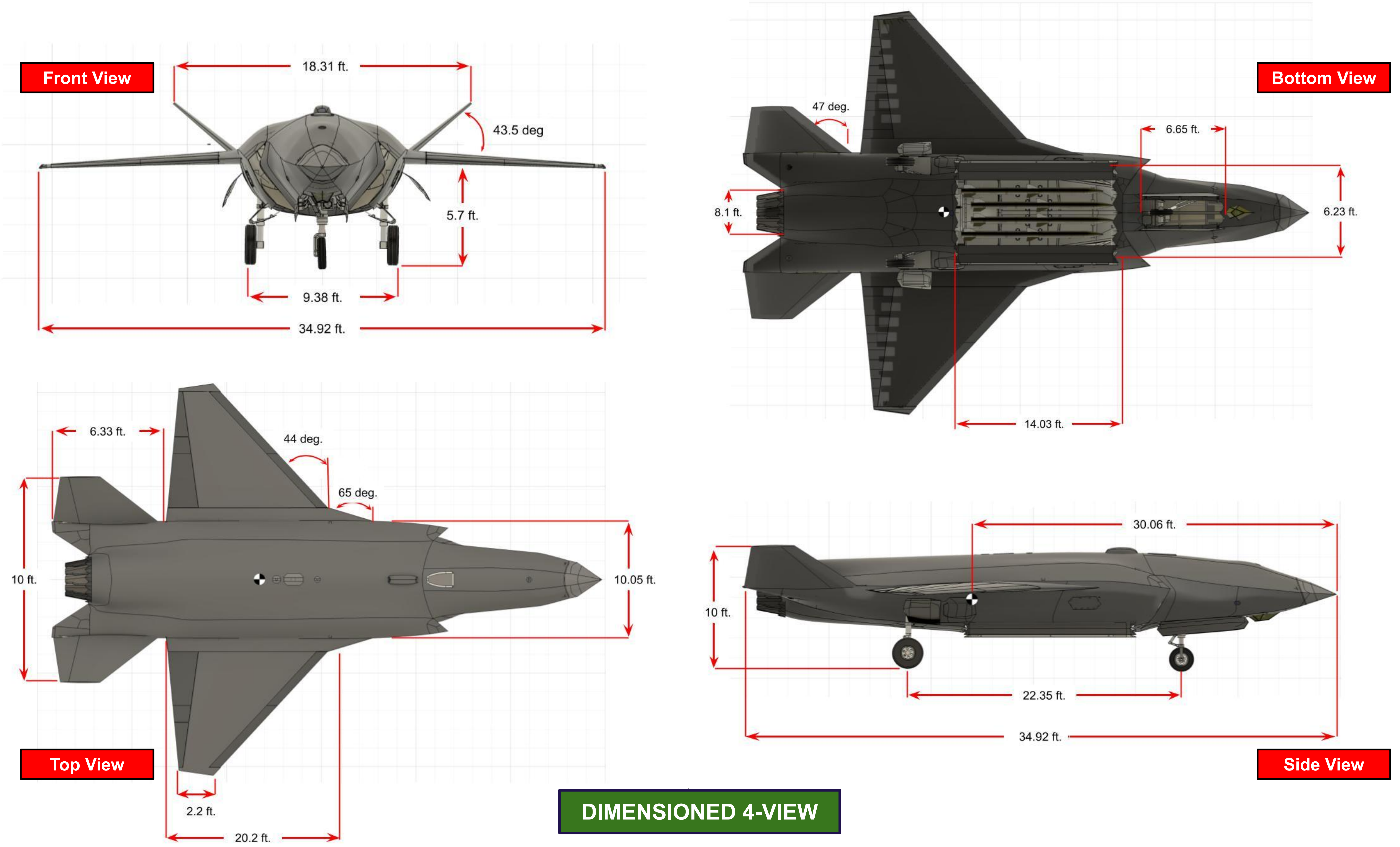
In the event of an enemy combatant being detected, the control hub is directly notified and appropriate air assets are able to be quickly sent for interception due to proper maintenance and drilling. During initial taxi and climb, the XQ-37 will be controlled by the GCS via direct LOS communications. As the aircraft reaches its operational altitude and begins to move toward its next objective, communication will be passed to BLOS capabilities, supplemented with satellite infrastructure. In carrying out the mission, the XQ-37 can operate on its own, with supporting communication infrastructure to allow GCS control. This includes ample internal fuel stores to complete the most constraining Combat Air Patrol mission, the capability to carry both long and short range missile ordinance, and redundant onboard communication and control systems to maintain remote controllability. However, system integration is a key focus for current and future systems acquired by the Department of Defense, and by designing our aircraft with this in mind has allowed for extension of the XQ-37's capabilities when properly supported. This includes a refueling probe to extend range, and in the case of a satellite communication denied environment, direct control can be facilitated either with supporting AWACS or 5th generation fighter aircraft, where direct control can be passed with the help of onboard MADL hardware.

Upon reaching interception range of the enemy combatants, the F2T2EA kill chain will commence. The first step is to find the target, which was already completed with the assistance of the early detection infrastructure. Next the XQ-37 must be able to fix onto the target, which is done using the onboard AESA radar in conjunction with EOTS to provide a visible identification in excess of 40 nautical miles. Upon fixing onto the target, these same systems will work with the onboard targeting computer to provide a weapon fix. In performing the targeting step, no lethal actions will be done without direct human authorization. Therefore a direct communication link back to a party with the proper authorizations to engage with lethal force will be required. Upon the granting of this authorization, the XQ-37 will be cleared to engage, releasing appropriate munitions either from the internal weapon bay or from underslung pylons on the wings. After engaging the threat, the kill chain will close with a final assessment of the vicinity, confirming whether the engagement resulted in a kill, along with searching for additional threats, in which case the kill chain will restart. This final step is fulfilled via the onboard tracking hardware, with supplemental information garnered by the existing intelligence infrastructure supporting the mission.

After the kill chain has concluded, the XQ-37 will return to base. During operation, additional considerations were made related to the remote pilot capabilities of the aircraft. This includes incorporating "soft-autonomous" autopilot behavior, to increase the survivability of the aircraft in the event of loss of communication with the aircraft. This includes the ability to return to base and maintain a static holding pattern while communications are attempted to be regained with the aircraft. These capabilities are supplemented with internal systems including LiDAR and INS sensors to be able to better interpret the aircraft's position in space. Additionally, there is an internal kill command to deny the enemy of gaining intelligence via compromised drones. Upon finally landing at a capable airbase, proper maintenance and review procedures commence between the airframe and crews to continue any further operations.

2.2 Proposed Solution

The 4-view layout of the XQ-37 with dimensions is provided below. The location of the CG is pictured and located from the nose of the aircraft.



Front View
18.31 ft.
43.5 deg
5.7 ft.
9.38 ft.
34.92 ft.
Bottom View
47 deg.
6.65 ft.
8.1 ft.
6.23 ft.
14.03 ft.
Top View
6.33 ft.
44 deg.
65 deg.
10 ft.
10.05 ft.
2.2 ft.
20.2 ft.
Side View
30.06 ft.
10 ft.
22.35 ft.
34.92 ft.
DIMENSIONED 4-VIEW

Table 1: Key Parameters for the XQ-37 Reaver

Parameter	Value
Flyaway Cost	$24.68M
Aircraft Empty Weight	15565 (lbs)
Mission Payload Weight (4 AIM - 120 AMRAAM)	1308 (lbs)
Maximum Payload Weight	5500 (lbs)
Maximum TOGW	25973 (lbs)
Top Speed	1.6 (Mach)
Instantaneous Turn Rate	18 (deg/s)
Time to Climb (35k ft)	58 (s)
Distance to Climb (35k ft)	4.8. (nmi.)
Endurance for CAP Mission	4 (hrs)
Intercept Range (Combat Mission)	215 (nmi.)
Intercept Range (Escort MissiOn)	185 (nmi.)

2.3 Driving Design Requirement I

2.3.1 ConOps Communication Requirement

Figure 2.1 shows the operational environment of the XQ-37 as it communicates with other aircraft as well as ground and satellite stations.

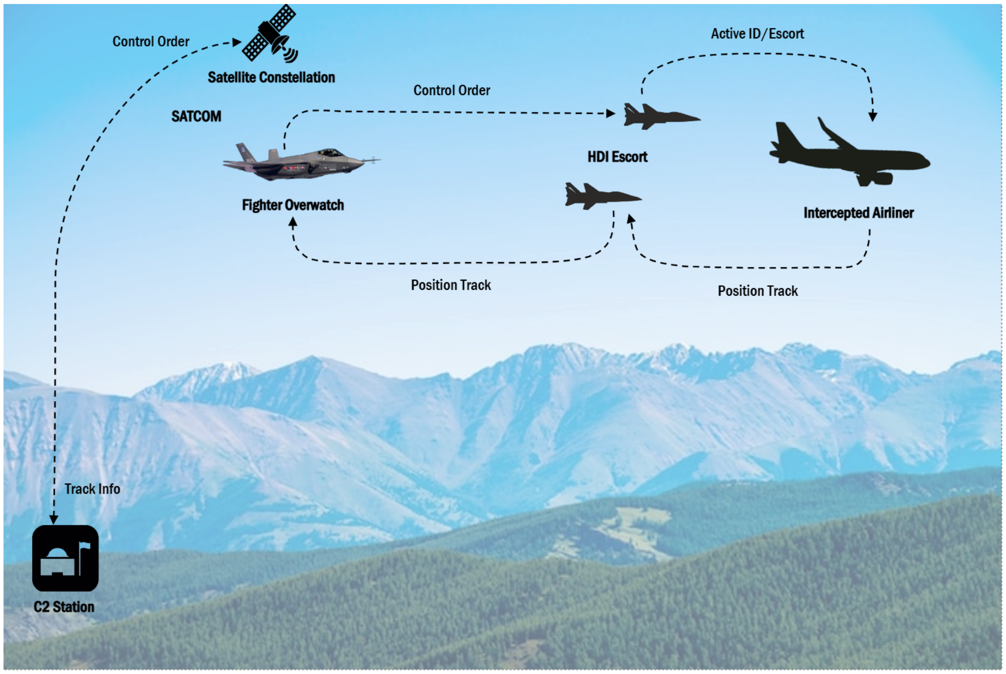


Figure 2.1: OV1 - Communication Requirement

Although not explicitly stated in the Request for Proposal, our team placed significant emphasis on requirements surrounding communication systems in order to ensure mission continuity and operational safety. In the event of a communications failure, the aircraft must possess multiple reliable backup systems capable of maintaining data links with the remote pilot to prevent mission failure and/or public safety risks. To address this, we developed a calculated and specific set of requirements aimed at maintaining consistent control of the aircraft. These include: full redundancy for all mission-critical subsystems, including mission critical communication systems, support for both Line-of-Sight and Beyond-Line-of-Sight communication modes, latency not exceeding 100 milliseconds, bandwidth above 100 Mbps, air-to-air communication interoperability, a design framework with accommodations for future upgrades, and a pre-programmed return-to-base or kill-chain protocol that activates after a predefined period of lost communication. These requirements, if effectively met, drastically mitigate the potential for mission failure, loss of aircraft, and threat to public safety.

2.3.2 Necessary Data Link Systems

In order to meet these communication requirements, a series of various data link protocols were investigated. The most promising of such protocols were compiled in Table 2, seen below. Table 2 shows the most useful information about each data link system. This includes; whether they can function in line-of-sight (LOS) or beyond line-of-sight (BLOS), their range, latency, and bandwidth. Due to the military nature of many of these systems, up-to-date information was often limited, so many of these values are based off of publicly released numbers or estimations.

The first data link protocol in Table 2, is the tactical common data link (TCDL). The Link 16 protocol, a typical military standard, is a TCDL which functions within line of sight through the use of radio frequencies to transmit information. They allow low-latency communication, with the necessary bandwidth, but are limited in range, due to the nature of a line-of-sight connection.

The next data link protocol present in Table 2, is the multifunction advanced data link (MADL). The MADL was developed primarily for the Lockheed Martin F-35 Lightning II for in-air communications between aircraft. It is a secure, low latency, stealth-centered protocol, but is even more limited in range than the traditional tactical common data link. Multifunction advanced data link systems are also restricted to communicating only with other MADL-equipped aircraft. That means additional data links are required for any other less advanced aircraft.

Tropospheric scatter, or troposcatter, was also analyzed in Table 2. It is a beyond line-of-sight form of communication that uses high-powered radio waves sent into the troposphere of the Earth. From there, the waves will refract and bounce back towards the surface, scattering as they do. It allows for further communication range, but requires high-powered equipment to send signals and large antennas to receive the signals. Due to the function of the signals refracting and scattering, the majority of the signal strength does not end up at the receiver, making for an inefficient form of communication.

The next data link protocol, in Table 2, uses Starshield's low Earth orbit (LEO) satellite constellations to provide secure beyond line-of-sight communication. Starshield is a subsidiary of SpaceX, whose satellite constellations are designed specifically for military use. The range will span the globe, with much lower latency than current satellite communication systems due to the lower altitude that the constellations orbit in, with some as low as 200 miles. Although the signal must travel further through the constellation, the latency is still greater. It also provides ample bandwidth to support our aircraft.

The most traditional military protocol for beyond line-of-sight communication is satellite communications, or SATCOM as seen in Table 2. It employs satellites typically orbiting in GEO, geostationary Earth orbit, at altitudes greater than 22,000 miles. It also allows for communications around the globe, but due to the greater altitude, this method of communication suffers from increased latency, making remote drone piloting difficult.

The final data link system present on Table 2, is the active electronically scanned array, or AESA, radar, which is more of a back-up system, should the main radio systems fail. It can be operated in line-of-sight communications, with an impressive range, latency, and bandwidth, but has limited use case since communication is not the radar's primary function.

Table 2: Data Link Protocols

Data Link	LOS/BLOS	Range *	Latency *	Bandwidth *
Tactical Common Data Link (TCDL)	LOS	~175 nm	1 – 10 ms	275 Mbps
Multifunction Advanced Data Link (MADL)	LOS	< 50 nm	1 – 10 ms	~100 Mbps
Troposcatter	BLOS	~400 nm	50 – 100 ms	50 – 200 Mbps
LEO Starshield	BLOS	3000+ nm	20 – 50 ms	220 Mbps
SATCOM	BLOS	3000+ nm	500 – 700 ms	150 Mbps
AESA (limited)	LOS	~300 nm	1 – 10 ms	550 Mbps

*Limited or Restricted Information

Of these analyzed data link protocols, four were chosen to fulfill all of the homeland defense interceptor's (HDI) communications requirements. These four are TCDL, MADL, LEO Communications, and SATCOM. Figure 2.2 below shows how these systems interact with the XQ-37. For operations within line-of-sight, the aircraft can use the tactical common data link, TCDL, to communicate with the command and control, C2, station. For any situation where the aircraft is beyond line-of-sight, it can employ satellite communications through the SATCOM or LEO communication pathways. The signal will be taken from the aircraft and relayed through the satellites back to the C2 station. The XQ-37 will also be outfitted with a multifunctional advanced data link, MADL, to facilitate secure air to air communications with other MADL-equipped aircraft, like the F-35 Lightning II.

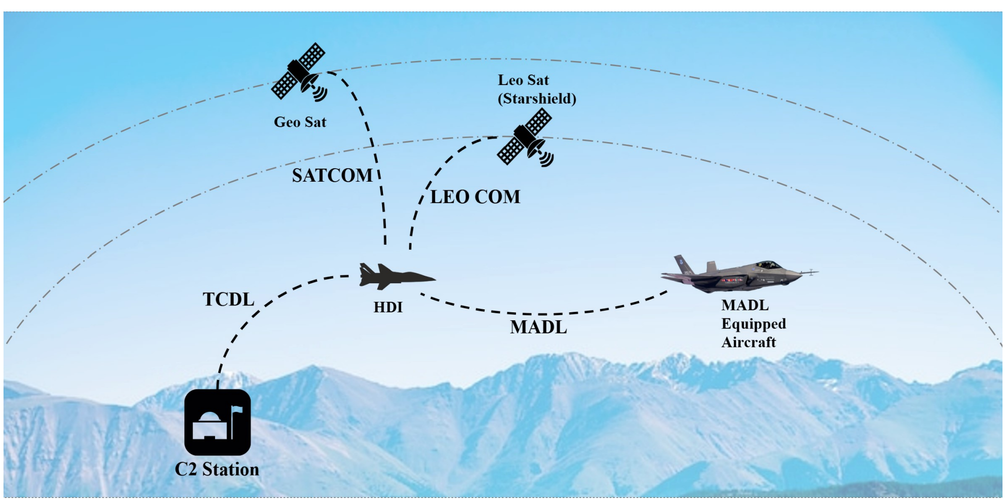


Figure 2.2: Data Link Interaction

2.3.3 Radio System Selection

Three different military aircraft radio systems were analyzed to see which could accommodate the necessary data links, as shown in Table 3 below. Those three listed systems are the multifunctional information distribution system (MIDS) - Joint Tactical Radio System (JTRS), the MIDS - low volume terminal (LVT), and the AN/ARC-210 radio communication system. These radio systems are listed in order of modernity, with the MIDS-JTRS being the newest radio system. This table shows that the MIDS-JTRS is the only radio system that can accommodate the new data links that the XQ-37 will utilize. The MIDS-JTRS is a software based system, with three additional waveforms left available. This lets new communication signals to be input and with the use of updated software, future data links can be added. This is how the Starshield LEO constellation communications will be incorporated. Additionally, this allows the system to be upgradeable, with more waveforms available to be used as more advanced data links become available.

Table 3: Radio Systems

Radio System	Cost	Capabilities				Future Proof
		Link 16	MADL	SATCOM	LEO Starshield	
MIDS - JTRS	$250 - 400k	☑	☑	☑	☑	Software Based with 3 additional waveforms
MIDS - LVT	$175 - 250k	☑	☒	☑	☒	Hardware Based with no additional waveforms
AN/ARC - 210	$45 - 75k	☒*	☒	☒	☒	Not Upgradeable

*Only Link 11

2.3.4 Communication Architecture

The XQ-37 uses a series of arrays and interface devices to utilize all of these data link protocols. The overall communication architecture can be seen in Figure 2.3 below. The tactical common data link (TCDL) can be accessed directly through the MIDS-JTRS' onboard radio system. Additionaly, the TCDL can be used through the AESA radar in case of a failure with the MIDS-JTRS radio. The multifunctional advanced data link (MADL) functions over four antenna array assemblies that are strategically placed across the XQ-37 to provide full 360 degree coverage. The antenna array assemblies can also be used to receive satellite communications over SATCOM. The MADL signal from the antenna array assemblies must then go through the MADL antenna interface unit to convert it to a signal recognizable by the MIDS-JTRS. There are also phased array antennas primarily for satellite communications, both from SATCOM and LEO communications. The LEO communications must first be processed through the LEO modem, to be used by the MIDS-JTRS. The MIDS-JTRS is connected directly to the aircraft's flight computer, so that any communication information retrieved can be used during the mission.

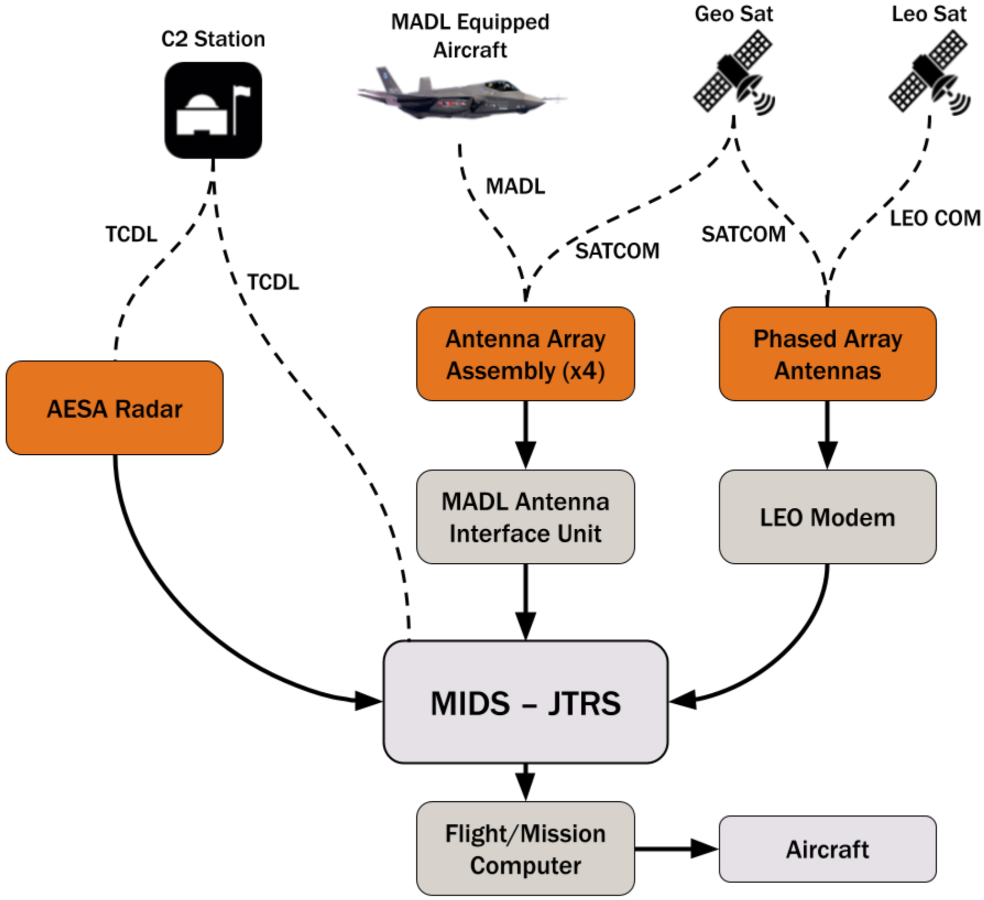


Figure 2.3: Aircraft Communication Architecture

2.4 Driving Design Requirement II

2.4.1 ConOps Loiter Requirement

Figure 2.4 shows the operational environment for a combat air patrol situation, where the XQ-37 loiters until intercepting enemy combatants.

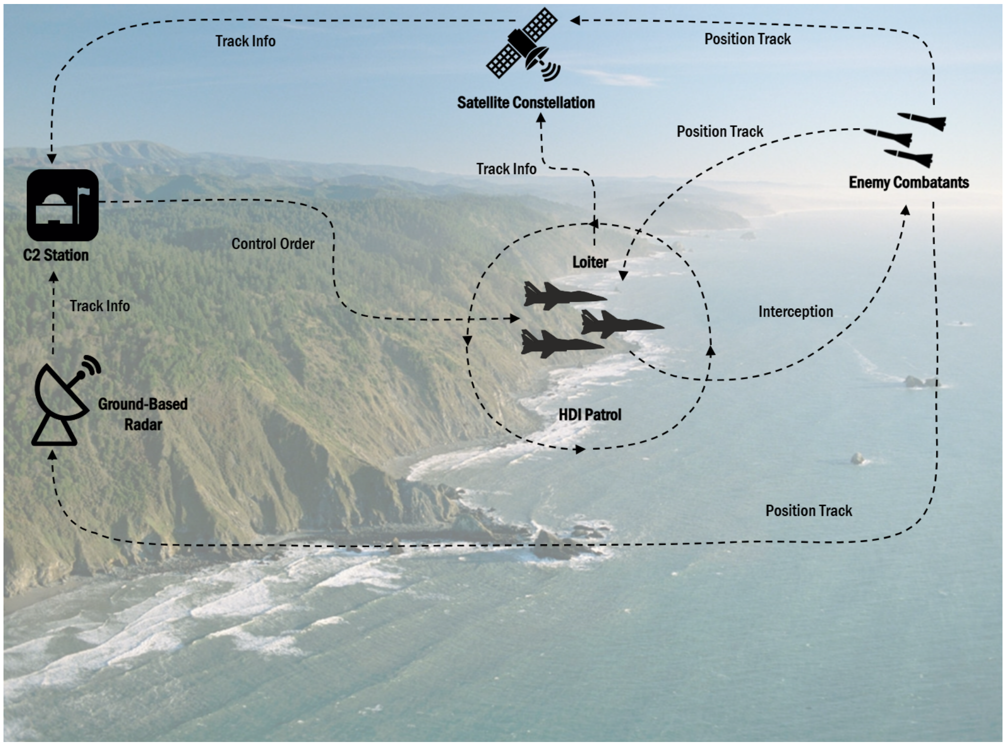


Figure 2.4: OV1 - Loiter Requirement

The RFP requires that the proposed solution completes all mission segments, but the four-hour loiter in the Combat Air Patrol mission proved to be the driving mission segment in the design process for the XQ-37. The size of the aircraft was initially increased due to the fuel required to complete this mission segment, driving wing planform area and maximum lift-to-drag ratio refinements. The aircraft must also have a minimum factor of safety of 1.5, pushing collaboration between the structures team and the performance team to ensure both adequate wing box stress and fuel storage. As other major design refinements were made to the XQ-37 to improve stability, such as pushing the wing back and the transition to a V-tail configuration, loiter performance maintained a priority for the team.

2.4.2 Fuel Allocation

As the combined mission segments from the Combat Air Patrol mission, specifically the loiter segment, require a large amount of fuel, fuel tank space was prioritized in the aircraft.

Utilizing the detailed CAD Model with the internal arrangement of the fuselage frames finalized, fuel tanks were then fitted within the fuselage of the XQ-37 in order to determine the overall design of the fuel tanks as well as their placement and fitment within the airframe. Within the fuselage of the XQ-37 are 9 fuel tanks, contoured to fit within the bulkheads as well as the outer mold line of the aircraft. Fuel was also allocated to be carried within the wings in a wet wing configuration, meaning fuel is free to move from one spar section to another, as there are no firewalls or bladder to store fuel within the wings. In total, 1950 gallons of fuel are allocated in the fuselage, and an additional 170 gallons are stored in the wings, for a combined total of 2120 gallons, or 13,500 lbs of fuel. However, without the ability to further optimize weight distribution due to the constraint of time, all fuel tanks were selectively and partially filled to 9100 lbs, which is used for all missions.

In addition, the XQ-37 also has the capability to refuel in-air, through its top mounted receptacle designed to operate with the USAF refueling tankers. If desired, the XQ-37 can also go through a conversion process to be outfitted with a refueling probe to work with the US. Navy's drogue and chute system.

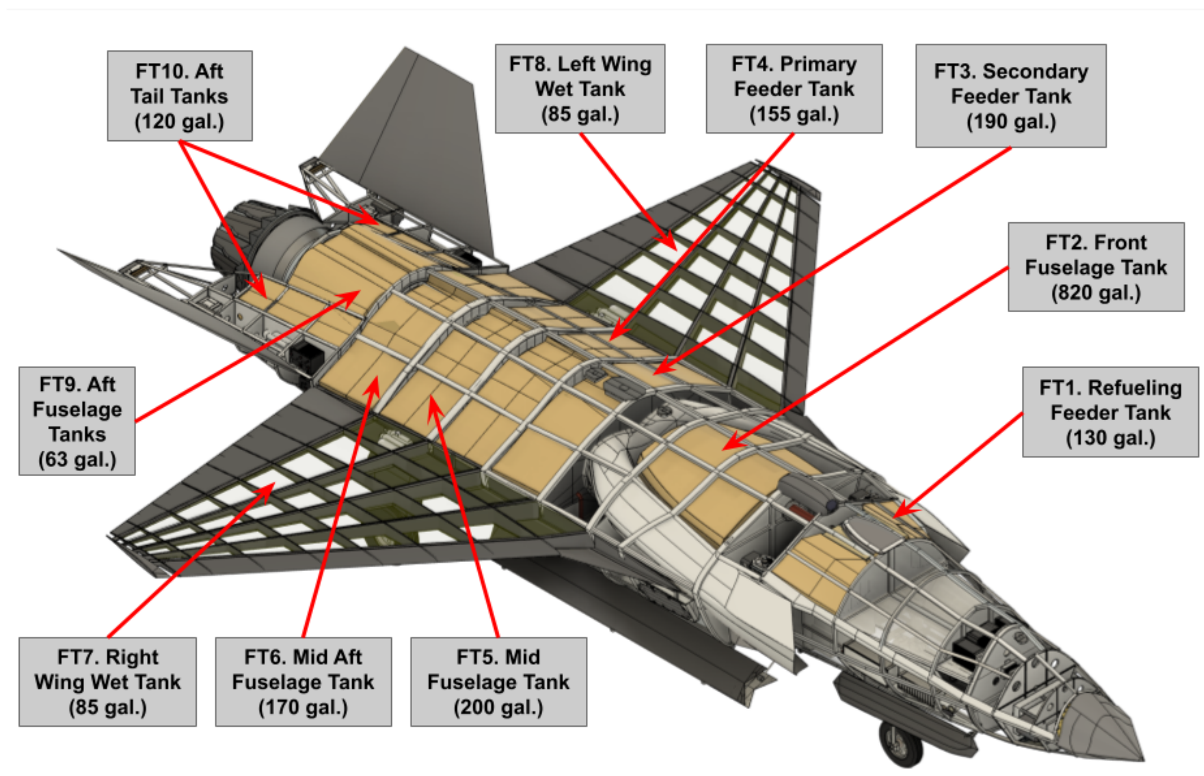


Figure 2.5: Fuel Tanks Layout and Capacity

2.4.3 Aerodynamics

The loiter requirement pushed a series of aerodynamic design decisions to maximize endurance while meeting performance metrics. Despite the large fuel allocation, the fuel burned in the Combat Air Patrol mission had to be less than 10,000 pounds to meet performance metrics. In addition to endurance, the configuration had stability and stealth requirements that pushed its design.

The foundation of the wing design are the planform area and the aspect ratio. Preliminary constraint analysis showed that wing loading would be dictated by the instantaneous turn rate requirement of of 18° per second which can be seen in Figure 2.6. With updated weight and balance figures, performance calculations show that a wing loading of 70.7 lb/ft^2 or lower will meet this requirement. As a result, our planform area has to be at least 368 ft^2.

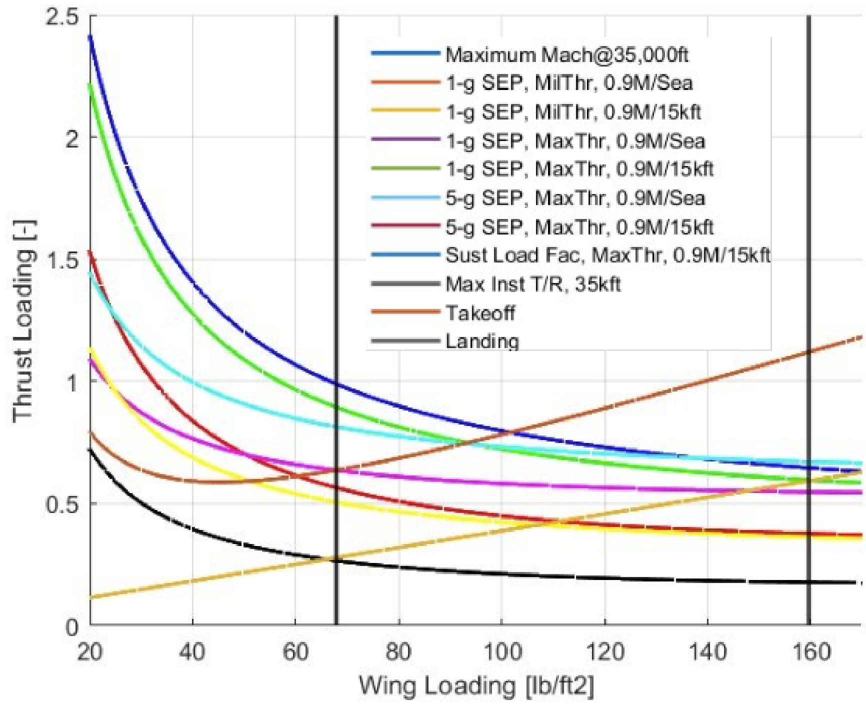


Figure 2.6: Preliminary Constraint Analysis

Wing aspect ratio is dictated by the Combat air Patrol mission. For efficient fuel burn, our preliminary lift over drag goal in endurance conditions was 13.5. VSPaero was used to perform a trade study on wing loading and aspect ratios effect on lift over drag, which can be seen in figure 2.7. Despite the study showing that we need an aspect ratio of 4 or greater to meet a L/D of 13.5, by reducing the wing loading below 70 lb/ft^2 and reducing our L/D goal to 12.5, an aspect ratio of 3 or greater was shown to meet our requirements.

One of the primary drivers of the wing shape was the decision to have a triangular wing similar to the F-35, which can be seen in Figure 2.8. This wing shape reduces radar cross section which improves stealth and survivability. A high aspect ratio is inhibited by a triangular wing because it focuses the bulk of the wing closer to the aircraft's body. In addition, updated weight and balance estimates were showing a center of gravity very close to the tail of the aircraft. This meant that the wings had to be shifted back for a neutrally stable static margin. Due to this shift, the wing would have to overlap with a conventional tail. As a consequence, the decision was made to change to a v-tail configuration. With a v-tail, the wings could slightly overlap the tail, and the center of gravity shifted forward enough to move the wings forward as well. This configuration had the added benefit of reduced cost and weight.

After the decision to switch to a v-tail configuration we were able to size our wing. An aspect ratio of 3.056 and a planform area of 377.74 ft^2 result in a wing loading of 69.44, and an instantaneous turn rate of 18.16° per second. Component buildup method in conjunction with VSPaero gave us a resulting L/D of 12.35 at endurance conditions.

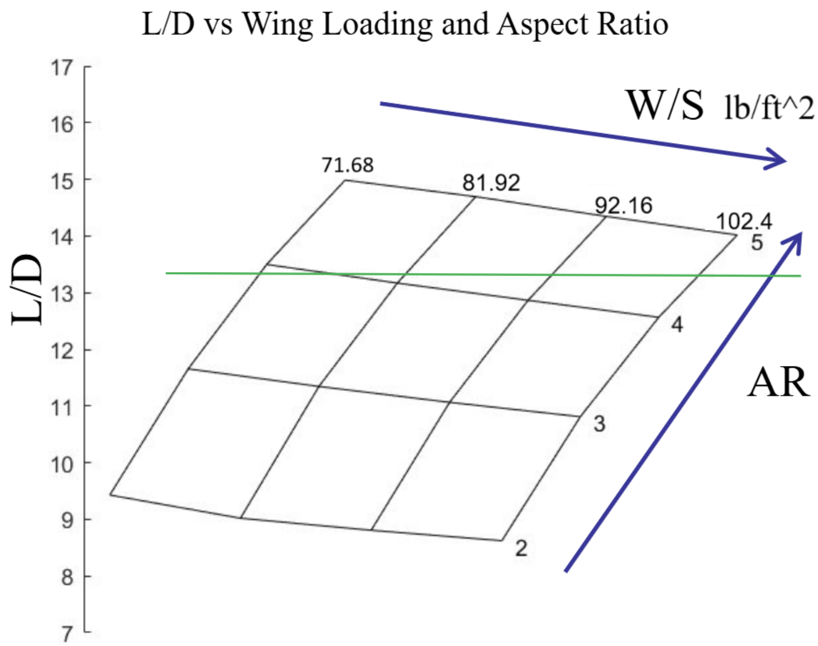


Figure 2.7: Preliminary Lift over Drag Trade Study

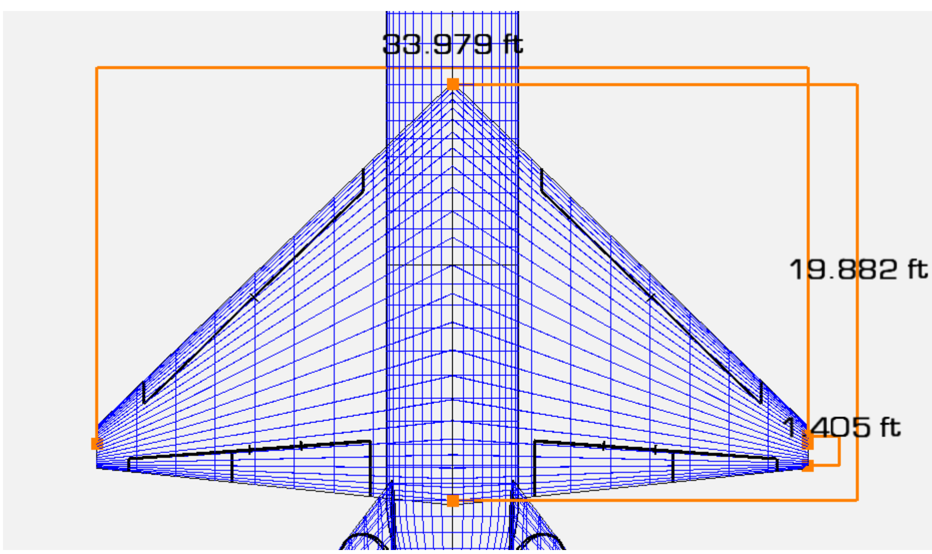


Figure 2.8: XQ-007 Wing Planform

2.4.4 Wing Box Design

With the wing geometry finalized and the control surfaces sized, a stress and buckling analysis of the wing box was performed to determine a preliminary placement of the front and rear spars, as well as the rib spacing. In addition, this analysis determined the required thicknesses of the ribs, spars, and skin. The wing box was approximated as a rectangular structure for analytical simplicity, serving as a conservative estimate. Constant rib spacing and uniform thickness distributions were assumed.

More specifically, separate stress analyzes were performed at the wing root for each of the four critical flight conditions identified from the V-n diagram, seen below in Figure 2.9. This diagram defines the aircraft's structural flight envelope, ranging from a negative load factor of -3 to a positive load factor of 7, as specified in the RFP. The four corners of the V-n diagram correspond to the critical combinations of load factors and airspeeds, representing the maximum positive and negative load factors that impose the greatest structural demands on the wing. This approach ensures that the wing structure is capable of withstanding the full range of load factors, operating speeds, and structural forces defined by the flight envelope. A factor of safety of 1.5 was applied across all conditions as required by the RFP. This accounts for uncertainties and ensures robustness in the preliminary design. More information on the stress analysis performed on the wing box can be seen in the Supplementary Information section of this proposal.

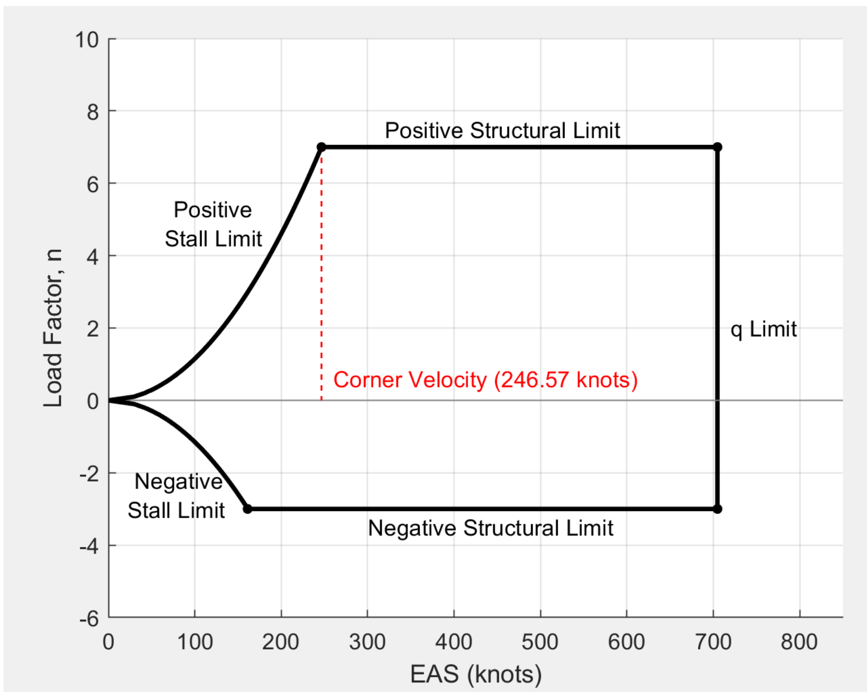


Figure 2.9: Vn Diagram at 35,000 ft MSL

In addition to the stress analysis, a buckling analysis was performed along the wingspan to ensure the wing box structure can withstand compressive and shear forces. This analysis focused on the wing skin and spars, which are key components in maintaining the wing's strength and stability during flight. The goal of the buckling analysis was to determine the critical forces that could cause the skin or spars to buckle. The calculations took into account factors such as the thickness of the skin, the spacing of the ribs, and the overall dimensions of the wing. Further details on the buckling analysis can be found in the Supplementary Information section of this proposal. This buckling analysis, in addition to the stress analysis, provides confidence that the wing will perform safely and effectively during flight.

The final wing box layout, which can be seen below in Figure 2.10, was defined by the results of the stress and buckling analysis. The placement of the spars and the spacing of the ribs were chosen to meet structural requirements while keeping the design efficient and lightweight. More specifically, the final wing box design includes five C-channel spars and ten ribs, with rib profiles following a NACA 64A-205 airfoil at the root and transitioning to a NACA 64A406 at the tip. The front spar is located where the leading edge slats hinge and the rear spar is located where the flaps/ailerons start. The ribs are spaced 10 inches apart. The wing skin has a uniform thickness of 0.05 inches. The wings are designed as wet wings, integrating fuel storage directly into the sealed wing box structure, with a capacity of up to 1,400 lbs of fuel across four bays per wing. A more detailed description of this wing box design can be found in the Supplementary Information section.

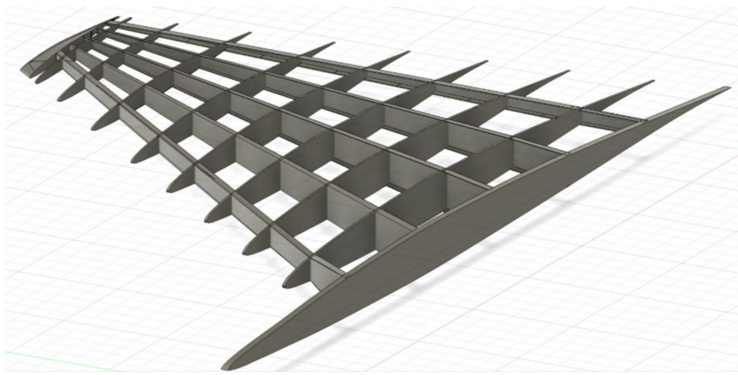


Figure 2.10: Wing Box Design

Materials for each wing box component were chosen based on structural role, strength, weight, and cost. The spars are made of 7075-T6 aluminum for its high strength and fatigue resistance at a relatively low cost. The ribs use 2024-T3 aluminum, offering a good strength-to-weight ratio. The wing skin is composed of IM7/8552 graphite epoxy composite, chosen for its excellent strength and fatigue performance while minimizing weight. Overall, this preliminary wing box design provides sufficient internal volume for fuel storage while meeting structural requirements, ensuring it can withstand critical flight loads without risk of buckling or yielding.

2.4.5 Propulsion

The Defensive Counter-Air Patrol Mission drove the design of the inlet, specifically the capture area. As this is the most constraining mission for the XQ-37 Reaver, the decision was made to optimize the inlet size for the loiter segment. The sizing of the capture area was performed using Equations 15.1 and 15.2 from Nicolai, which are listed below as Equation 1 and Equation 2 respectively [3].

$$\frac{\dot{m}_\infty}{\dot{m}_c} = \frac{\rho_\infty V_\infty A_\infty}{\rho_\infty V_\infty A_c} = \frac{A_\infty}{A_c} \tag{1}$$

$$A_{\infty_E} = \frac{\dot{m}_E + \dot{m}_S}{32.17\rho_\infty V_\infty} \tag{2}$$

Standard atmosphere values were obtained using the *atmosisa* function in MATLAB. Secondary airflow from engine oil cooling, hydraulic system cooling, and ejector nozzle was assumed. The secondary airflow values were obtained from Nicolai Table 15.1, assuming standard values [3].

With these assumptions, the inlet capture area was designed to be 3.81 feet squared per inlet for a dual-inlet design. Other mission segments will require smaller areas to achieve engine demand flow rate, thus bypass doors may be needed for excess air. In the detailed design phase of the XQ-37 Reaver, a Diverterless Supersonic Inlet will be implemented to minimize boundary layer air ingestion. At this stage of the design, the duct has only been preliminarily shaped to hide the compressor face of the engine.

2.4.6 Stability and Control

Figure 2.11, below, shows the CG travel during the Defensive Counter-Air Patrol Mission, which is the constraining mission for the XQ-37 Reaver. Two fuel burn strategies are compared: an optimized case (blue) and a non-optimized case (green). Fuel is strategically drawn from multiple tanks per mission segment to minimize the variation in CG travel during flight in the optimized case. The shift in CG for this case is less than 0.1 feet forward. The non-optimized case drains one fuel tank at a time, which causes the CG to steadily shift almost a full foot aft as fuel burns. The comparison clearly shows how fuel distribution plays a significant role in keeping the aircraft stable during flight.

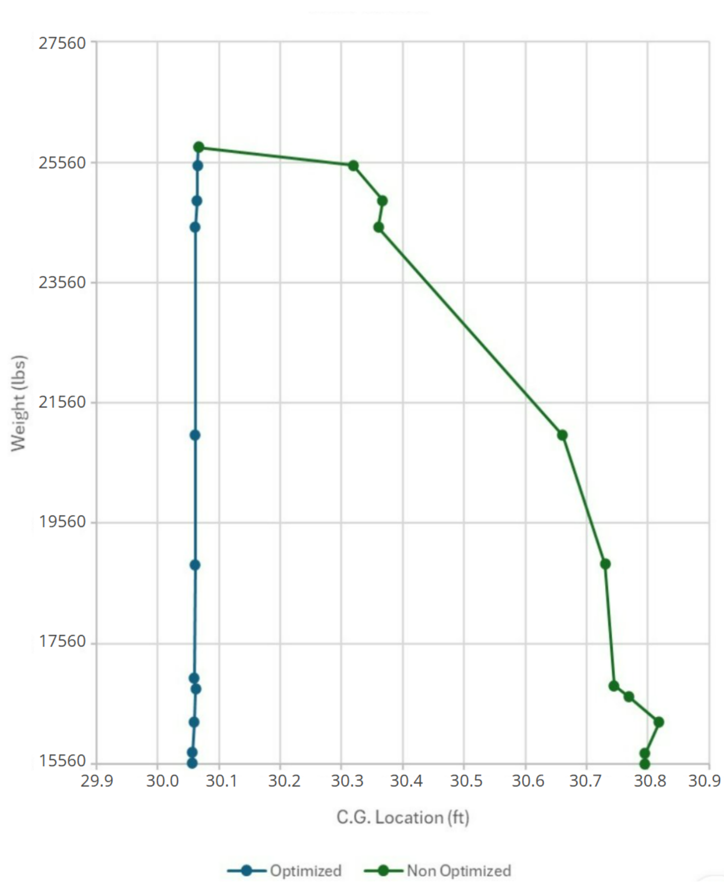


Figure 2.11: CG Travel Plot - Defensive Counter Air Patrol Mission

The aircraft was designed to meet the SM requirement even in the non-optimized case, since stability is vital for an aircraft of this nature. The following table gives the SM (at subsonic speeds) for different sections of the mission segment for the non-optimized case.

Table 4: Static Margin at Each Mission Segment

Mission Segment	Avg. CG Location (ft)	SM
Nominal	30.06	0.91%
Takeoff-Climb-Cruise	30.20	-0.35%
Loiter	30.50	-3.05%
Combat	30.70	-4.85%
Landing	30.80	-5.75%

The XQ-37 Reaver employs a V-tail configuration that was chosen to reduce aft weight compared to an earlier twin-tail stabilator design. This design change was made to resolve a challenge in regards to the aircraft's CG location caused by large central fuel tanks (needed to meet the loiter requirement) and the removal of a M61A1 20 mm Cannon (removed to allow space for sensitive avionics in the nose). In addition to solving the CG issue, the V-tail design reduces complexity and cost by decreasing weight and the number of actuators required.

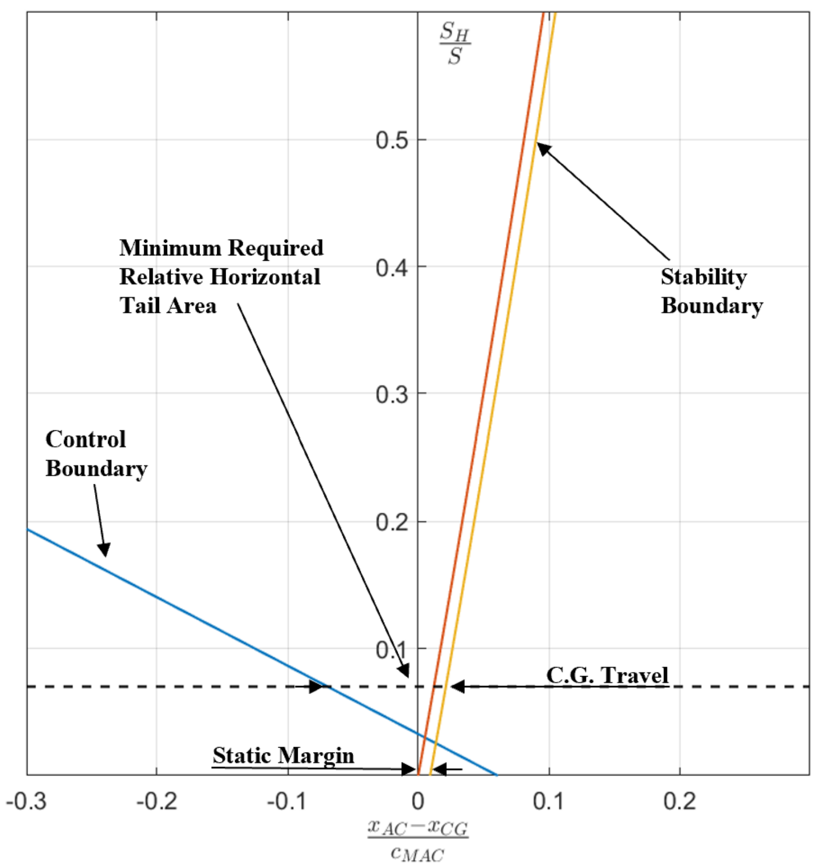


Figure 2.12: Scissor Plot used for Tail Sizing

The XQ-37 has a triple-redundant flight controller that is used to ensure optimal mission performance and resolve any potential instability issues. The XQ-37 is naturally both statically and dynamically stable with the optimized fuel distribution system, so the flight controller serves as a redundant system. Additionally, the flight controller facilitates the implementation of a return-to-home feature in the case of communication loss and reduces pilot workload by allowing for an increased stability mode for long loiter and cruise segments and a lower stability mode for combat segments where increased maneuverability is desired. In regards to stability and control, the XQ-37 is fine tuned for optimal mission performance.

2.5 Conclusion

From the inception of this program, resilient communication systems were one of Cyberdyne's top priorities. It is paramount that a remotely piloted system of this caliber must remain reliably connected across any airspace at all times, as any lapse in communication poses a significant operational and public safety risk. To guarantee proper mitigation of this risk, the XQ-37 utilizes redundant mission-critical systems, ensuring survivability in the event of signal loss or hardware malfunction. It operates on both LOS and BLOS communication channels, with sub-100ms latency for responsive control, and over 100 Mbps bandwidth for minimal lag time between operator and aircraft with high-fidelity ISR data streaming across the entirety of the national domain. Its seamless air-to-air interoperability allows for coordinated fleet operations with advanced fifth generation fighters, expanding its utility across all national security and allied contexts. Moreover, this platform has been engineered with modularity to support future upgrades, ensuring long-term relevance in a rapidly evolving technological environment.

The loitering requirement, which at first proved to be a strenuous challenge within the aircraft's cost and performance constraints ultimately shaped one of its most valuable capabilities. Through strategic design trade-offs and methodical optimization of nearly all aspects of each technical domain, the XQ-37 emerged as a high-endurance, practical platform with unparallelled mission flexibility. This requirement transformed the aircraft into a multi-role asset capable of extended presence utilizing high quality surveillance, rapid response lethality, and a significant strategic reach across a wide range of missions over the continental United States and its allies.

In satisfying these two crucial requirements, the XQ-37 has evolved into a reliable, robust, and highly capable platform optimized for modern homeland defense by balancing affordability with lethality, autonomy with control, and flexibility with mission success.

Cyberdyne Aerospace is proud to present the XQ-37 Reaver, a strategic force multiplier for the United States, purpose-built to secure the homeland, and ready to serve as a cornerstone of America's aerial defense infrastructure.

3 Supplementary Information

3.1 Requirements

The requirements were tracked using the Mathworks requirements toolbox. As seen in Figure 3.1 and Figure 3.2 the XQ-37 Reaver meets all requirements as set by the team.

requirements*				
1	#2	Icy Landing Distance		
2	#63	Cost		
3	5	Manufacturing		
4	3	Structures		
5	#44	Aircraft Systems		
6	#45	Mission Systems		
7	#46	Mission Systems		
8	#47	Mission Systems		
9	#48	Mission Systems		
10	#49	Mission Systems		
11	#50	Mission Systems		
12	#51	Mission System		
13	#52	Performance		
14	#53	Performance		
15	#54	Performance		
16	#55	Performance		
17	#56	Performance		
18	#57	Performance		
19	#58	Performance		
20	4	Structures		
21	5	Structures		
22	6	Structures		
23	7	Stability and Control		
24	8	Mission Performance		
25	9	Mission Performance		
26	10	Performance		
27	12	Aircraft Systems		
28	13	Aircraft Systems		
29	14	Project Deadline		
30	17	Payload		
31	18	Stability and Control		
32	19	Stability and Control		
33	20	Stability and Control		
34	22	Stability and Control		
35	23	Stability and Control		
36	24	Stability and Control		
37	25	Stability and Control		
38	26	Stability and Control		
39	27	Stability and Control		
40	28	Mission Systems		
41	29	Performance		

Figure 3.1: Requirements Table Part 1

42	30	Mission Systems		
43	31	Mission Systems		
44	32	Mission Systems		
45	33	Mission Systems		
46	34	Mission Systems		
47	35	Safety		
48	36	Aerodynamics		
49	37	ConOps		
50	39	ConOps		
51	40	ConOps		
52	41	ConOps		
53	#42	Aircraft Systems		
54	#43	Aircraft Systems		
55	#59	Performance		
56	#60	Performance		
57	#61	Performance		

Figure 3.2: Requirements Table Part 2

3.2 ConOps

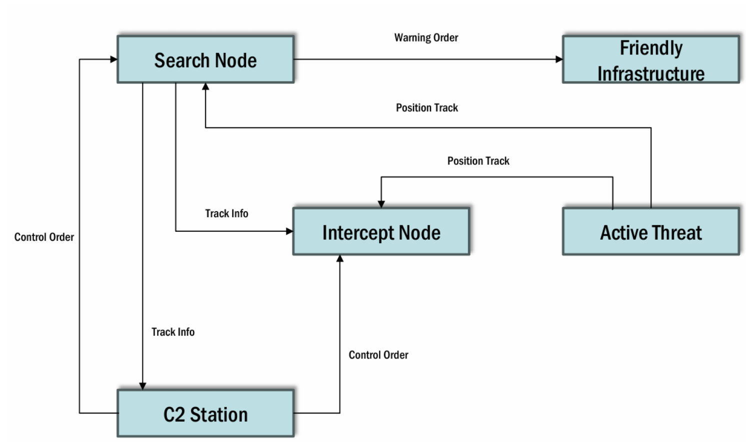


Figure 3.3: OV-2 Diagram for Operation of XQ-37

The OV-2 diagram, shown in Figure 3.3, is meant to illustrate the resource needs and exchange patterns between components used during the operation of the XQ-37. The two main nodes which contribute to the completion of the mission are the Search and Intercept Nodes. The Search node is composed of systems and infrastructure which are able to detect possible threats from long ranges.

It uses this info to inform both military and civilian assets to take the proper precautions. The Intercept Node is composed of the XQ-37 aircraft which is sent into the direct vicinity of the active threat to use its onboard systems to properly track and engage the threat. Both these nodes are coordinated and directed by the C2S, which provides control orders which are determined after interpreting inbound track info on the current situation.

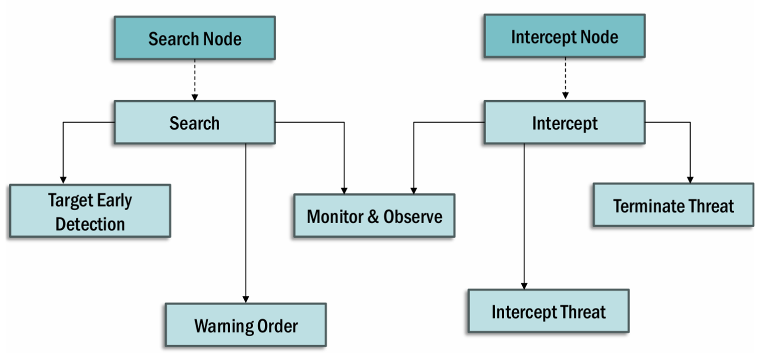


Figure 3.4: OV-5 Diagram for Operation of XQ-37

The OV-5 diagram, shown in Figure 3.4, is meant to illustrate the specific operations carried about by the various systems involved throughout the mission. The diagram is read left to right to show the general progression of the mission, again split between the two Search and Intercept Nodes, which support each other throughout the mission. The search node provides intelligence throughout the entirety of the mission. It provides the early detection of the target, initiating the start of the mission, leading to a warning order to affected parties including friendly infrastructure which could be targeted and the military assets housing the XQ-37 and supporting drone operators. Throughout the operation, the Search node continues to monitor and observe the detected targets, providing continual operational awareness updates. Meanwhile, the intercept carries out the physical interception mission objectives. This includes flying out to meet the threat in its immediate vicinity, and terminating the threat using its onboard systems. Meanwhile, the XQ-37 also carries out monitoring and observation operations with its internal systems, such as its AESA radar and EOTS imaging system. This facilitates its ability to carry out the kill chain.

3.3 Mission Profiles

The XQ-37 was designed to perform three unique missions as required by the AIAA 2025 RFP: a Defensive Counter-Air Patrol mission, a Point Defense Intercept Mission, and a Intercept/Escort mission. Of these three, the Defensive Counter-Air Patrol mission proved to be the most difficult mission to meet given the tight cost and performance constraints, and was the driving mission that ended up sizing our aircraft to optimize this platform. A summarized depiction of the three missions can be seen below in Figure 3.5.

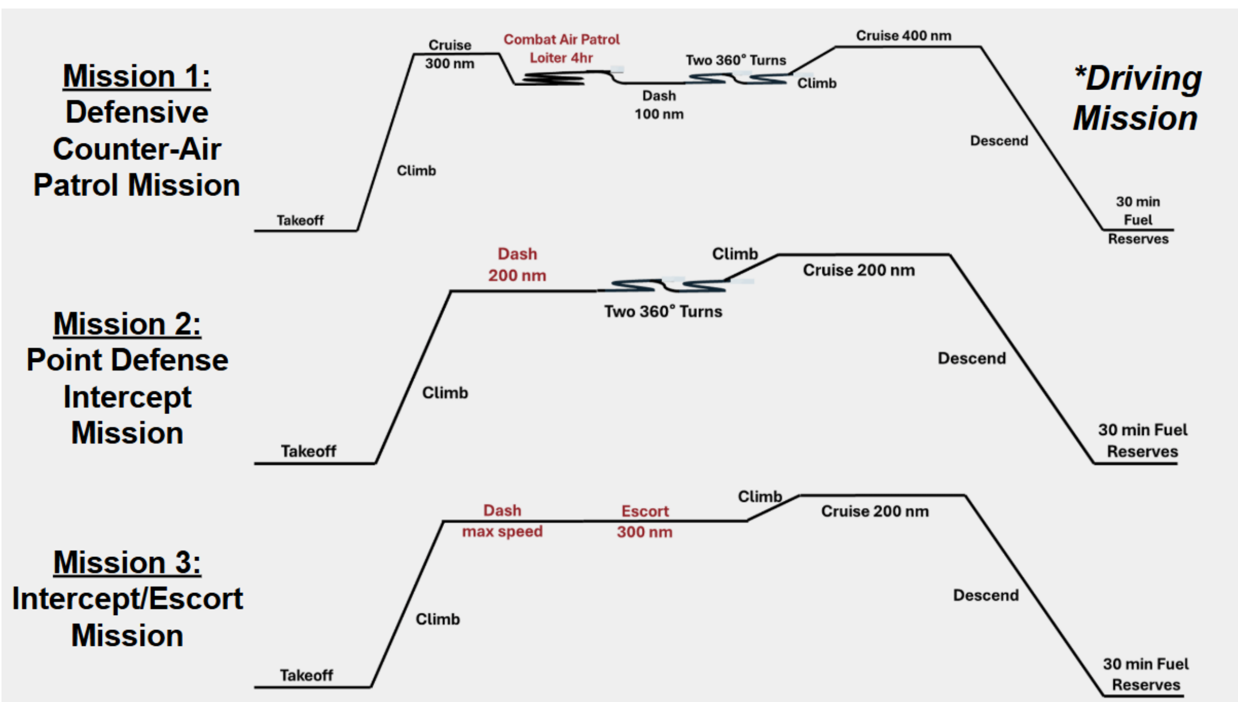


Figure 3.5: Mission Profiles Required by 2025 AIAA RFP

3.4 Comparator Analysis

The primary comparator aircraft for the mission profile defined by the RFP is the Lockheed Martin F-16C. Figure 3.6 shows the performance of our proposed solution, the XQ-37, compared to that of the F-16C. All values listed for the F-16C are directly from the United States Air Force fact sheet for the aircraft. Note that the cost listed in Figure 3.6 has been estimated based on the cost of 18.8 million dollars (1998) given on the fact sheet [4].

Aircraft	XQ-37	F-16C
Cost ($2024)	$24.68 million	$37 million
Max Takeoff Weight	25,973 lbs	37,500 lbs
Max Speed	Mach 1.6	Mach 2
Range (ferry)	4,500 nm	1,740 nm
Payload	1,308 lbs	5,426 lbs
Service Ceiling	68,250 ft	50,000 ft

Figure 3.6: XQ-37 Compared to the F-16C

The service ceiling of the XQ-37 was determined from the maximum rate of climb at altitude, and represents the absolute service ceiling of the aircraft. The overall ferry range of the XQ-37 is much better than its competitor, as the F-16 has an internal fuel capacity of 7000 pounds with a higher TOGW while the XQ-37 has an internal fuel capacity of 9100 pounds at a much lower TOGW [4]. The cost of the XQ-37 Reaver meets the RFP requirement of 25 million dollars, whereas the F-16C is noncompliant. Despite the lower payload of the XQ-37, the affordability and range capability make it a superior choice over the F-16C.

3.5 Measures of Merit

The primary measures of merit considered in the design process include range, time-to-climb, and minimum runway length required. As a Homeland Defense Interceptor, exceeding the requirement for minimum runway length can allow for a more flexible operating environment. The XQ-37 requires a total field length of 6245 feet, well below the required 8000 feet stated by the RFP. This field length includes operation in icy runway conditions, as the XQ-37 is equipped with a drag chute to minimize landing distance in such weather. With improved takeoff and landing performance, operation from a variety of airbases is made possible. This also provides more options in the event of an emergency.

The time-to-climb just after takeoff is essential for an interceptor-type aircraft. With a time-to-climb to 35000 feet of 58 seconds, the XQ-37 has a competitive advantage over most adversaries. This rate-of-climb is achieved with a high thrust-to-weight ratio, a driving factor in the selection of the aircraft's engine.

As the XQ-37 will often complete combat air patrol missions, range is an important factor when improving the design. Having a high range of 4500 nautical miles allows the aircraft to search for incoming threats for an extended duration without in-air refueling. The integration of in-air refueling further enhances the combat air patrol capability of the aircraft.

3.6 Promising Technologies

The United States Department of Defense, DoD, is currently pushing an initiative called Joint All-Domain Operations, or JADO. JADO is an initiative to integrate the capabilities and resources of the entirety of the DoD. [5] JADO consists of many different programs and initiatives for every branch of the military. One of these that is additionaly branch-wide is Joint All Domain Command and Control, JADC2. JADC2 is an initiative to connect all sensor information across every vehicle in the US military, as well as NATO allies, to facilitate more informed decision making. It will function by creating a network of networks to allow many different systems to interact with each other [6].

The Air Force's contribution to these initiatives is through the Advanced Battle Management System, ABMS. The ABMS comprises the USAF efforts to link together all of their sensor information, both across the Air Force and the Space Force [7]. In the future, this same interconnective technology will be utilized across the other military branches.

Another promising technology is the diverterless supersonic inlet, or DSI, that the XQ-37 uses. It is a new inlet design that can feed the engine clean air separated from boundary layers without using a splitter plate. It uses a bump on one side of the inlet to redirect the unwanted boundary layer airflow away from the intake. This results in a much simpler and lighter intake system and avoids additional costs for costly diverters. [8]

3.7 Aerodynamics

3.7.1 Airfoil Selection

Airfoil selection and our aerodynamics strategy in general were based on simplicity and proven systems. The mission of the XQ-37 has a similar profile to the F-16 and similar performance values, so our design was heavily inspired by it.

The F-16 uses a NACA 64A204 airfoil. We selected a slightly thicker NACA 64A205 a=0.8 mod airfoil at the root and a NACA 64A406 a=0.8 mod airfoil at the tip which can be seen in Figure 3.7 and Figure 3.8 respectively. 64A series airfoils were designed by NACA to remove the cusp at the end of a supercritical airfoil [9]. This airfoil shape is ideal for supersonic aerodynamics, structural design, and control surface integration.

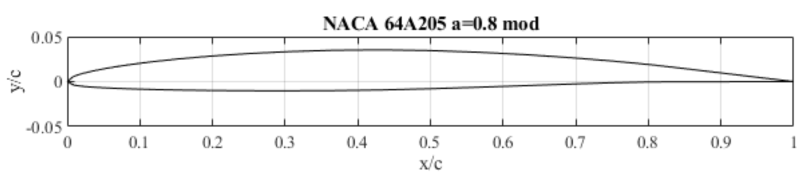


Figure 3.7: Root Airfoil

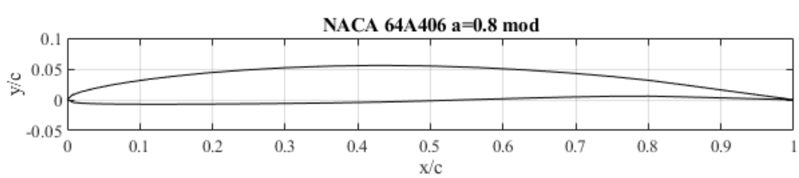


Figure 3.8: Tip Airfoil

Our lift curve was calculated using VSPaero and can be seen in Figure 3.9. The calculation was done using Vortex Lattice Method at Mach 0.350, a reference area of 377 ft^2 and the Carlson pressure correlation as a stall model. This exhibits stall at around 12 degrees of angle of attack,, which is consistent with experimental data from NACA on the similar NACA 64A210 airfoil [9].

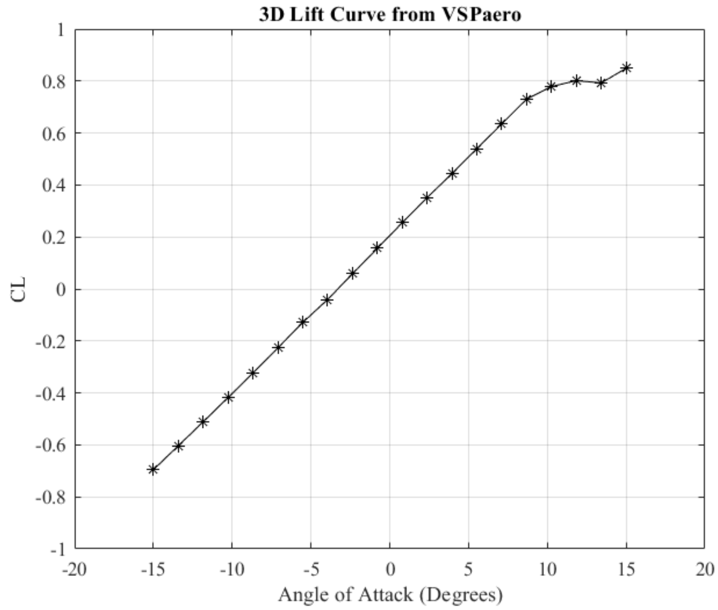


Figure 3.9: 3D Lift Curve From VSPaero

3.7.2 Wing Geometry

Our wing geometry is defined by its planform area, chord, taper, sweep, twist, and dihedral which can be seen in Table 5. Constraint analysis dictated that our aircraft required a certain wing loading in order to meet our most restrictive Combat Air Patrol mission. Loiter requirements in this mission dictated our wing loading and aspect ratio. Parameters were selected with considerations to the aircraft configuration, static margin, supersonic performance requirements, and stealth requirements.

Table 5: Planform Parameters

Wing Geometry Summary	
Planform Area	377.74 ft^2
Mean Aerodynamic Chord	11.11 ft
Aspect Ratio	3.056
Taper Ratio	0.1
Leading Edge Sweep	44°
Twist	-2°
Dihedral	-2°

Our wing has a triangular shape with a forward trailing edge sweep similar to the F-35 to improve stealth which can be seen in Figure 3.10. This shape in conjunction with our 44° leading edge sweep limited the aspect ratio. The wings also had to be shifted close to the tail due to static margin requirements. This led to a v-tail configuration to limit overlap between the wing and tail.

A trade study was performed to help decide the Aspect Ratio and Wing Loading of our aircraft based on the resulting Lift over Drag which can be seen in Figure 3.11. The green horizontal line shows the goal L/D which is based on our endurance requirements. Our resulting planform of 377.74 ft^2 and aspect ratio of 3.056 meet this requirement with the triangle wing shape.

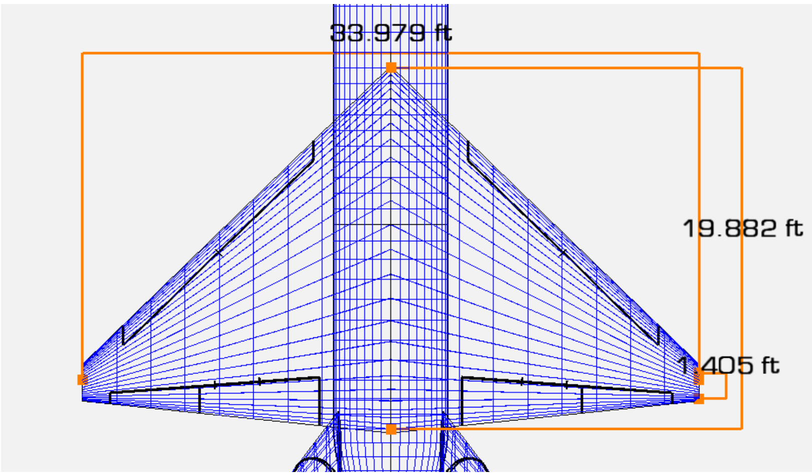


Figure 3.10: Wing Planform Shape

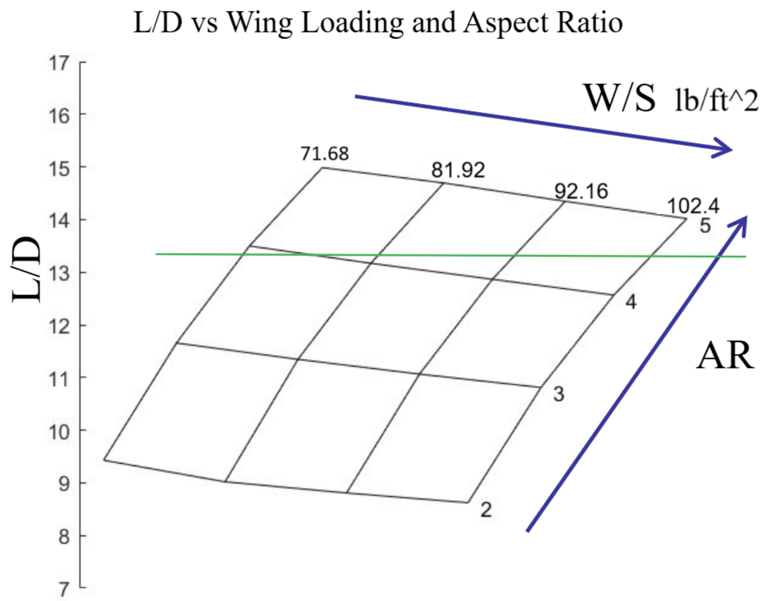


Figure 3.11: Lift over Drag Trade Study

A dihedral of -2°results in relaxed roll stability and higher responsiveness in roll. The -2° twist causes our wing to stall closest to the root of the wing before the tip which allows for more control during a stall. The leading edge sweep of 44° reduces wave drag which is discussed in more detail in section 3.7.4.

3.7.3 High Lift Devices

The high lift device configuration was chosen based on simplicity and sized based on performance requirements. The configuration is highly inspired by the F-16, with plain flaps, and leading edge slats [10]. The v-tail required the flaperons of the F-16 to be split into flaps and ailerons. Flaps and ailerons assist as high lift devices during takeoff by increasing the wing camber, and the leading edge slats prevent stall at high angle of attack by reducing flow separation. During high angle of attack maneuvers, the leading edge slats will align with the freestream creating a less aggressive leading edge.

Takeoff and landing requirements dictated that the aircraft would require a coefficient of lift of 1.3 at an angle of attack below the tip back angle of 18°. The flaps and ailerons were iteratively sized on VSPaero until this requirement was met, which resulted in flaps and ailerons depth of 0.16 times the chord. The lift curves of resulting from this analysis can be seen in Figure 3.12. We meet our takeoff condition at 13° angle of attack and a flap plus aileron deflection of 25°.

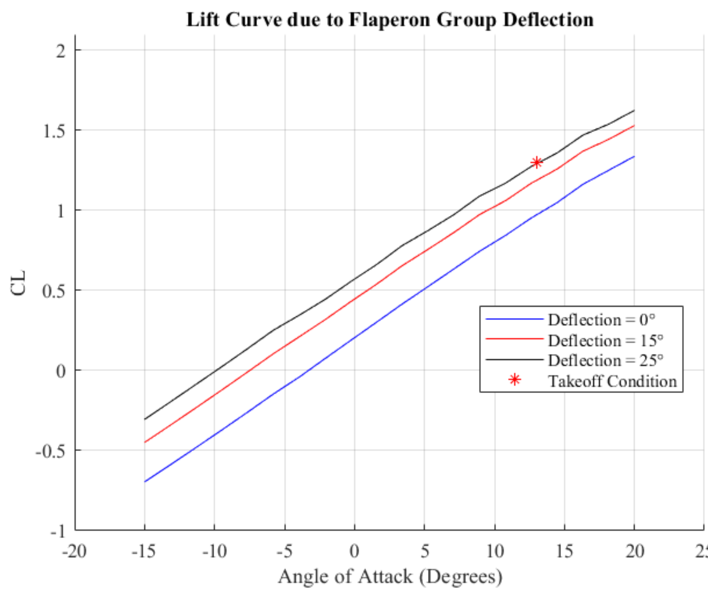


Figure 3.12: Lift Curves at Takeoff and Landing Conditions

3.7.4 Drag Buildup

The total drag buildup for the XQ-007 consists of zero-lift drag, induced drag, and wave drag. Zero-Lift drag was calculated using the component buildup method described by HAW-Hamburg [11]. The component buildup method calculates zero lift drag for each component with consideration to skin friction drag, pressure drag, and interference drag.

Table 6: Zero-Lift Drag Buildup

Component Drag Buildup			
Mach	0.85	1.2	1.6
Wing	0.003136	0.00299	0.00278
Tail	0.00334	0.00318	0.002955
Fuselage	0.00275	0.00247	0.002193
External Stores	0.00324	0.00291	0.002568
Total	0.01248	0.01155	0.010496

Induced drag was calculated using VSPaero at different mach conditions. The Karman-Tsien Mach Correction was used in this calculation, and only the induced drag was taken from VSPaero due to the low accuracy of wave drag and zero-lift drag from the software. At each mach condition, the aircraft was trimmed to level flight at 35,000 ft. The coefficient of lift decreases as the Mach number increases.

Table 7: Total Drag Buildup at 35,000 ft

Total Drag Counts				
	Induced	Viscous	Wave	Total
Mach 0.85	0.01448	0.01248	0.000000	0.02695
Mach 1.2	0.01805	0.01155	0.022243	0.05184
Mach 1.6	0.01075	0.0105	0.019508	0.04076

Wave drag was calculated using an empirical relations described by Brandt [12]. These equations can be seen below.

$$C_{D_{\text{wave}}} = \frac{4.5\pi}{S}\left(\frac{A_{\max}}{l}\right)^2 E_{\text{WD}}(0.74 + 0.37\cos\Lambda_{\text{LE}})\left[1 - 0.3\sqrt{M - M_{C_{D_0}\max}}\right] \tag{3}$$

$$M_{C_{D_0}\max} = \frac{1}{\cos^{0.2}\Lambda_{\text{LE}}} \tag{4}$$

$$M_{\text{crit}} = 1.0 - 0.065 \left(100 \frac{t^{\text{max}}}{c}\right)^{0.6} \cos^{0.6} \Lambda_{\text{LE}} \tag{5}$$

The wave drag equation Eq.3 requires calculations of critical mach number, which is the speed where wave drag starts to take effect, and maximum drag mach number. These values were calulated using Eq.eq:mcrit and Eq.(4) respectively. A high leading edge sweep increases the critical mach number and decreases the mach of maximum wave drag, which is why a high sweep of 44° was selected. Maximum cross sectional area and aircraft length are also factored into wave drag calculations using this method.

The empirical relations used to calculate wave drag did not consider the area ruling of our specific aircraft, and instead use a wave drag efficiency factor E_{WD} which was conservatively estimated at 2.2.

Drag polar was calculated at three different Mach numbers and can be seen in Figure 3.13. The drag polar was generated using VSPaero, but drag was refined by replacing zero lift drag from vsp from component buildup drag and wave drag. The resulting drag polar is consistent with historical aircraft designs described by Nicolai which can be seen in Figure 3.14 [3].

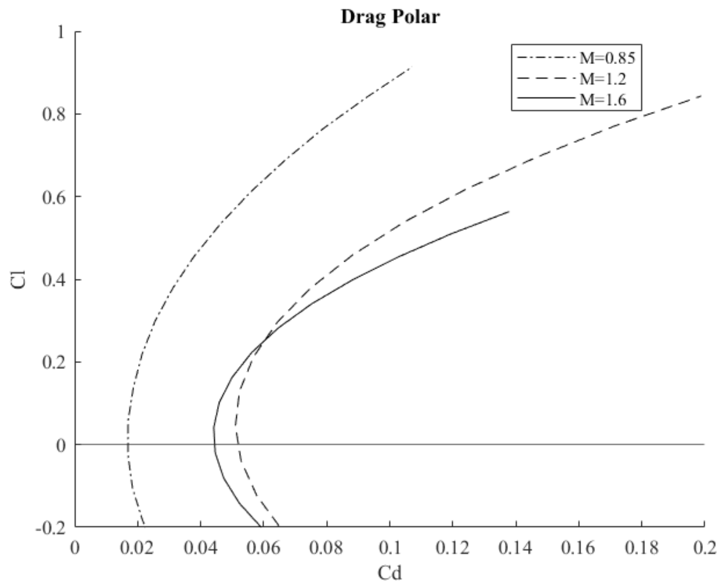


Figure 3.13: Drag Polar at Different Mach Conditions

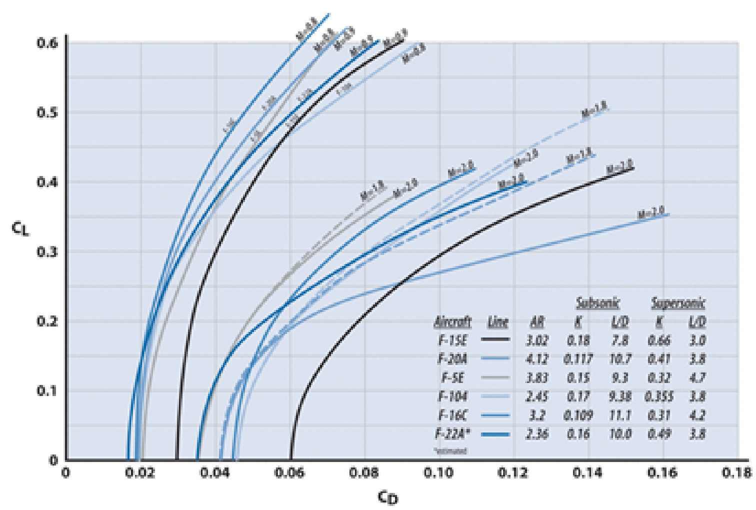


Figure 3.14: Historical Aircraft Drag Polar, Nicolai

3.7.5 Spanwise Lift Distribution

The spanwise lift distribution was calculated on VSPaero and can be seen in figure 3.15. The wing taper was selected with the goal of an elliptical lift distribution. The distribution from VSP is nearly identical to the approximated lift distribution used to design our wingbox which can be seen in figure 3.30.

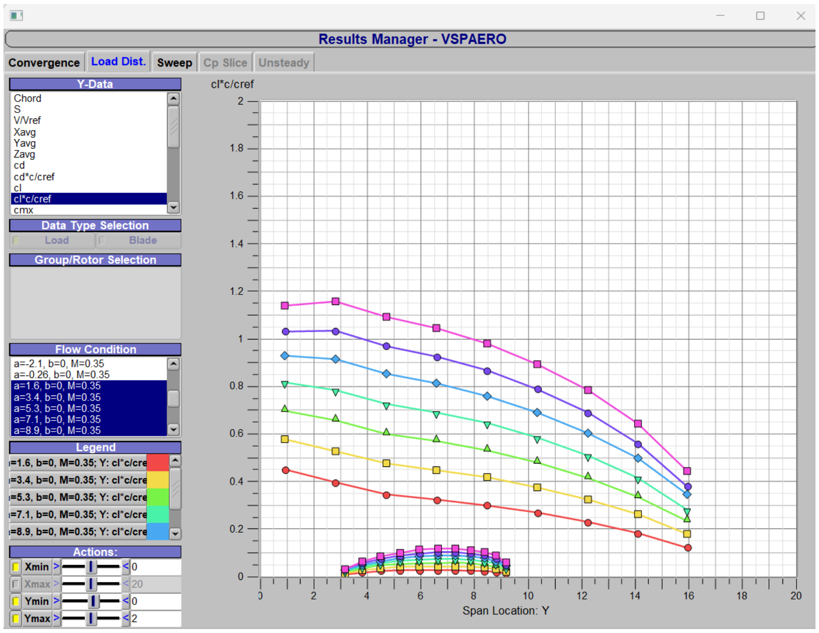


Figure 3.15: Spanwise Lift Distribution

3.8 Weight and Balance

3.8.1 Weight Breakdown

A detailed weight breakdown of the XQ-37 Reaver can be seen below in Table 8. Many of these values were determined using mass equations from historical data for fighter aircraft from Raymer's textbook [13]. The remaining are given in the RFP [14].

Table 8: Aircraft Component Weight Breakdown

Component	lb	Component	lb
Wing	2769.22457	Fuel Tank 4	77.8474003
Nose Frames	633.308398	Fuel Tank 5	99.3113648
Nose Gear Doors	107.032395	Fuel Tank 6	86.5692074
WPB/Inlet Frames	408.768857	Fuel Tank 7	55.928489
Engine Frames	97.4320613	Fuel Tank 8	96.0906691
Wing Box Frames	885.775427	Fuel Tank 9	108.489465
Tail Frames	365.37023	Fuel Tank 10	61.0546938
Weapons Bay Frames	192.079758	AIM 120	327
Weapons Bay Doors	360.232871	AIM 120	327
V Tail	303.83	AIM 120	327
Engine Housing	737.1814	AIM 120	327
Main Gear	700	Fuel - Tank 1	817.130091
Nose Gear	225	Fuel - Tank 2	2355.47305
Engine	3980	Fuel - Tank 3	844.370531
AESA Radar	450	Fuel - Tank 4	676.07469
Avionics & FC	1378	Fuel - Tank 5	862.480956
Fixed Equipement	873	Fuel - Tank 6	751.820226
IRSTS	50	Fuel - Tank 7	485.717388
Fuel Tank 1	94.0893872	Fuel - Tank 8	834.510454
Fuel Tank 2	271.223663	Fuel - Tank 9	942.189222
Fuel Tank 3	97.2260191	Fuel - Tank 10	530.236501

These mass estimates encompass various components, including the wing, engine housing, avionics and flight controller equipment, fixed equipment (such as electrical and hydraulics systems, and anti-icing), and the total fuselage weight. The total fuselage weight was further subdivided into the following sections: the nose frames, nose gear doors, WPB/inlet frames, engine frames, wing box frames, tail frame, weapons bay frames, and weapons bay doors. The individual mass values for these components were initially determined using the detailed CAD model by assigning material properties and densities. However, given that the CAD model is currently preliminary and does not incorporate a fully detailed structural layout, these calculated mass values were instead utilized as percentages of the total fuselage weight, which was determined through Raymer's established equations [13]. Fuel tank weights were determined using a similar approach. Each fuel tank was modeled within the CAD file to record its volume. The total fuel tank weight for the aircraft was established using Raymer's equations [13], and this total was then distributed among the individual tanks based on their respective volumes. The weight of the fuel itself was calculated based on the volume of each tank and converted to pounds using the density of JP-8 fuel [1].

A less detailed breakdown of the empty weight can also be seen below in Table 9. These values are all either given in the RFP [14], or determined using Raymer's equations [13]. The empty weight is broken down into three sections, the structures total, propulsion total, and the fixed equipment total. These sections add up to create a total empty weight of 15,563.06 pounds. As expected, the structural total makes up a vast majority of the empty weight, with the propulsion weight just slightly less due to the heavy engine.

Table 9: Empty Weight Breakdown

Component	lbs
WING	2769.22
V-TAIL	303.83
FUSELAGE	3050.00
MAIN LANDING GEAR:	700.00
NOSE LANDING GEAR:	225.00
ENGINE SUPPORTS:	737.18
STRUCTURE TOTAL:	**7785.23**
GE-F110-129 ENGINE	3980.00
FUEL TANKS/SYSTEM	1047.83
PROPULSION TOTAL:	**5027.83**
FLIGHT CONTROLS	730.00
ELECTRICAL & INSTRUMENTS	510.00
HYDRAULICS	160.00
AVIONICS	700.00
ANTI-ICING	150.00
AESA RADAR	450.00
IRSTS	50.00
FIXED EQUIPMENT TOTAL:	**2750.00**
EMPTY WEIGHT TOTAL:	**15563.06**

Finally, a weight breakdown can be seen below in Table 10. This displays the empty weight, which is made up of the basic structures and systems in the aircraft without the fuel or payload.

With a payload consisting of four AIM-120 AMRAAM missiles, as well as a total fuel weight of 9100 lbs, the maximum takeoff weight results as 25,973 lbs. This also accounts for an estimated taxi out fuel weight of 300 lbs, which is expected to be consumed before takeoff.

Table 10: Aircraft Weight Breakdown

Weight Breakdown	lbs
Empty Weight	15565
Payload 4 AIM - 120 AMRAAM	1308
Zero Fuel Weight	16873
Total Fuel Weight	9100
Ramp Gross Weight	25673
Taxi Out Fuel Weight	300
Maximum Takeoff Weight	25973

3.8.2 CG Calculation

The forward and aft center of gravity (CG) positions were determined based on the aircraft's full and empty weight conditions, respectively. Table 11 provides a detailed breakdown of the aircraft's component masses, along with the CG location of each component, measured in feet from the nose. Table 12, below, summarizes the total weight, moment, and resulting CG location for both the full and empty aircraft configurations. The full aircraft CG was calculated by summing the moments and weights of all components, including payload and fuel, and dividing the total moment by the total weight. The empty aircraft CG was determined using the same method, excluding payload and fuel. These CG values were used to help guide the placement of major structural components and fuel tanks to keep the CG within the static margin limitation to allow for safe flight. They define the aircraft's allowable loading envelope and ensure that stability and control are maintained throughout all phases of flight.

Table 11: Center of Gravity Components

Component	lb	ft	ft lb
Wing	2769.224573	32.4113	89754.1684
Nose Frames	633.3083982	8.942	5663.043697
Nose Gear Doors	107.0323955	11.013916	1178.845813
WPB/Inlet Frames	408.7688569	21.208	8669.169918
Engine Frames	97.43206127	38.815	3781.825458
Wing Box Frames	885.7754271	32.4113	28709.1331
Tail Frames	365.3702297	45.5	16624.34545
Weapons Bay Frames	192.0797578	25.25	4850.013885
Weapons Bay Doors	360.232871	25.25	9095.879993
V Tail	303.83	45.5	13824.265
Engine Housing	737.1814	41	30224.4374
Main Gear	700	34.25	23975
Nose Gear	225	41	9225
Engine	3980	12.5	49750
AESA Radar	450	7.06	3177
Avionics & FC	1378	12.5	17225
Fixed Equipement	873	20	17460
IRSTS	50	7.41	370.5
Fuel Tank 1	94.08938716	15.22285	1432.308627
Fuel Tank 2	271.2236627	22.459833	6091.63817
Fuel Tank 3	97.22601907	27.74295	2697.336586

Component	lb	ft	ft lb
Fuel Tank 4	77.84740028	30.843125	2401.057098
Fuel Tank 5	99.31136483	33.6895	3345.750225
Fuel Tank 6	86.56920741	36.62141667	3170.287015
Fuel Tank 7	55.92848904	40.381	2258.448316
Fuel Tank 8	96.09066907	43.313916	4162.063169
Fuel Tank 9	108.4894651	32.4113	3516.2846
Fuel Tank 10	61.05469376	38.815	2369.837938
AIM 120	327	25.25	8256.75
AIM 120	327	25.25	8256.75
AIM 120	327	25.25	8256.75
AIM 120	327	25.25	8256.75
Fuel - Tank 1	817.1300909	15.22285	12439.0488
Fuel - Tank 2	2355.473054	22.459833	52903.53144
Fuel - Tank 3	844.3705311	27.74295	23425.32943
Fuel - Tank 4	676.0746901	30.843125	20852.25618
Fuel - Tank 5	862.4809558	33.6895	29056.55216
Fuel - Tank 6	751.820226	36.62141667	27532.72176
Fuel - Tank 7	485.7173876	40.381	19613.75383
Fuel - Tank 8	834.5104536	43.313916	36145.91569
Fuel - Tank 9	942.1892218	32.4113	30537.57752
Fuel - Tank 10	530.2365013	38.815	20581.1298

Table 12: Center of Gravity Limits

	Weight (lbs)	Moment (lb ft)	C.G. (ft)
Full Aircraft	25972.06944	778134.9565	29.96045264
Empty Aircraft	15564.06633	472020.1399	30.327559

As mentioned above, Figure 2.11 below is the CG travel plot, or potato plot, for the Defensive Counter Air Patrol Mission. This plot consists of a optimized fuel burn case and a non-optimized fuel case. As discussed previously, the optimized case pulls fuel from multiple fuel tanks during each mission segment to minimize the CG travel throughout the mission to less than 0.1 foot. The non-optimized case pulls fuel from one tank at a time, resulting in CG travel of about 0.8 feet aft. This shift is directly impacted by the depletion of fuel in the tanks without regard to their location relative to the aircraft's reference datum. As the tanks towards the front of the aircraft are emptied first, the remaining fuel mass shifts backward, increasing the moment arm and shifting the CG aft.

To generate this plot, the aircraft component mass properties detailed in Table 11 above were used in addition to their respective longitudinal distances from the aircraft's nose. The point representing maximum weight on this plot corresponds to the aircraft at takeoff with a full fuel and payload load. As the mission progresses, the aircraft's total weight decreases primarily due to fuel consumption, calculated based on burn rates specific to each segment. A notable weight reduction and CG shift is during mission segment 4, the combat air patrol. This segment was a key design driver for our aircraft due its significant fuel consumption. Additionally, mission segment six, the combat phase, results in a significant weight reduction when all the missions on board are dropped. In the end, the plot visually shows the impact of fuel management on CG location throughout a mission, highlighting why an optimized fuel distribution strategy is important for maintaining the CG within acceptable flight envelope limits.

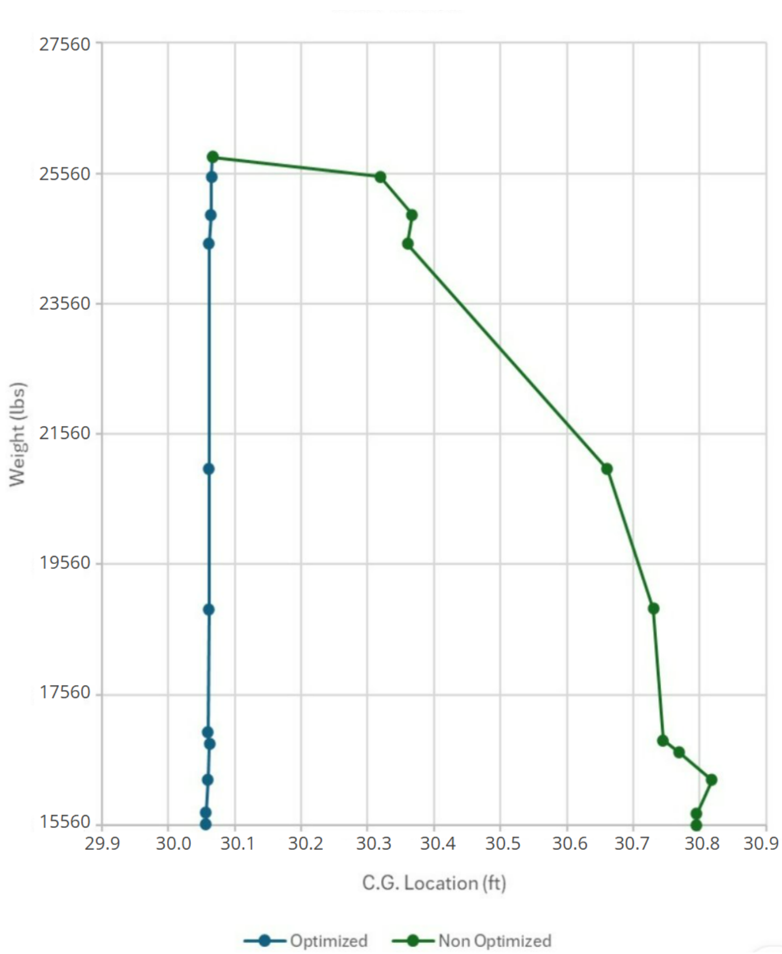


Figure 3.16: CG Travel Plot - Defensive Counter Air Patrol Mission

3.9 Stability and Control

3.9.1 Empennage and Control Surface Sizing

The XQ-37 Reaver employs a V-Tail configuration to reduce weight in the tail section and resolve CG issues. The V-Tail was sized based on the process outlined in Chapter 6 of Sadraey [15]. The horizontal projected area was initially sized based on the horizontal tail volume coefficient ($\bar{V}_H$) of similar aircraft and based on the l/L ratio. Next it was ensured that the horizontal projected area met the minimum required horizontal tail area as outlined by the scissor plot found in Figure 3.17.

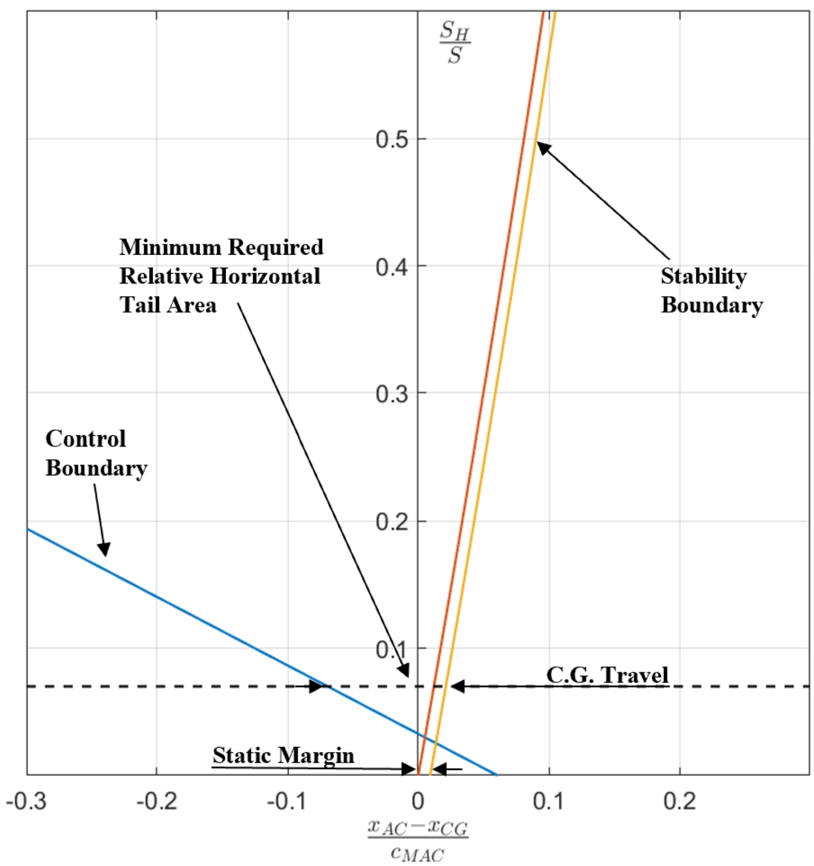


Figure 3.17: Scissor Plot used for Horizontal Tail Sizing

Similarly, the vertical projected area was initially sized based on the vertical tail volume coefficient ($\bar{V}_H$) of similar aircraft. Since the configuration is a V-Tail, the horizontal and vertical projected area are coupled, the effective area of the tails were able to be adjusted by manipulating the angle and size of the tails. To maintain a stealthy geometry, it was ensured that the tail surfaces did not form a 90 degree angle between the two surfaces. For high maneuverability and given the aircraft has an advanced flight control system, the V-Tail control surfaces are all moving. The V-Tail was iteratively fine tuned in VSPAero concurrent with static and dynamic stability analysis. The XQ-39 has ailerons that were initially sized based on historical data from similar aircraft and then verified and adjusted through dynamic stability analysis.

3.9.2 Static Stability

The longitudinal static stability was evaluated over the entire CG travel for the non-optimized fuel distribution system during the most constraining mission (most fuel so largest CG travel). Table 13 contains shows average SM for each mission segment of the Defensive Counter Air Patrol mission.

Table 13: Static Margin at Each Mission Segment

Mission Segment	Avg. CG Location (ft)	SM
Nominal	30.06	0.91%
Takeoff-Climb-Cruise	30.20	-0.35%
Loiter	30.50	-3.05%
Combat	30.70	-4.85%
Landing	30.80	-5.75%

Inspecting the table, the XQ-37 meets the $-10\% \leq SM \leq 10\%$ requirement stated by the RFP for all mission segments even with the non-optimized fuel distribution system. Lateral-directional static stability analysis was completed using VSPAero for the nominal CG location and the major findings are summarized in Table 14.

Table 14: Static Stability Analysis Results

Parameter	Value	Req. Met
SM	0.91%	✓
$C_{M\alpha}$	$-0.3283/rad$	✓
$C_{l\beta}$	$-0.7022/rad$	✓
$C_{n\beta}$	$0.0607/rad$	✓

Inspecting Table 14 the XQ-37 meets all static stability requirements set by the team derived from MIL-F-8785C [16].

3.9.3 Dynamic Stability

A preliminary dynamic stability analysis was completed using the moments of inertia from the CAD model, stability and control derivatives from VSPAero, and a linearized longitudinal and lateral-direction state matrices. The analysis was completed for the optimized fuel distribution system case, and the major findings are summarized in Table 15.

Table 15: Dynamic Stability Analysis Results

Parameter	Value	Req. Met
ζ_{ph}	0.050	✓
ζ_s	1.012	✓
T_R	$0.037s$	✓
ζ_{dr}	0.316	✓
T_{2s}	$22s$	✓

Inspecting Table 14 the XQ-37 meets all dynamic stability requirements set by the team derived from MIL-F-8785C [16].

3.10 Propulsion

3.10.1 Engine Selection

The RFP prioritizes a low-cost aircraft, thus the selection of a fixed engine is preferred to lower research and development costs. To select the best fixed engine for our aircraft, a trade study was performed with four engines. The four engines considered were F110-GE-129, F404 / RM12, F100-PW-229A and EJ2000 selected for their application in aircraft expected to have similar propulsion performance as the XQ-37. These engines are respectively installed in aircraft such as the F-16 Fighting Falcon, F/A-18 Hornet, F-15 Eagle, and the Eurofighter Typhoon [17][18][19][20]. The XQ-37 Reaver was initially sized with each of the four engines to compare the resulting TOGW. Figure 3.18 shows that the F110-129 manufactured by General Electric provides the lowest TOGW from initial sizing, around 27000 pounds. The TOGW shown in Table 10 from refined sizing provides a comparable value of 25973 pounds, confirming the validity of this trade study. As a lower TOGW is associated with a lower empty weight, and empty weight is a metric closely related to cost, the F110-129 is the best engine choice for the XQ-37. As this is the same engine installed in the F-16, the TOGW of the F-16 is comparatively shown to be 37500 pounds in Figure 3.18 [4].

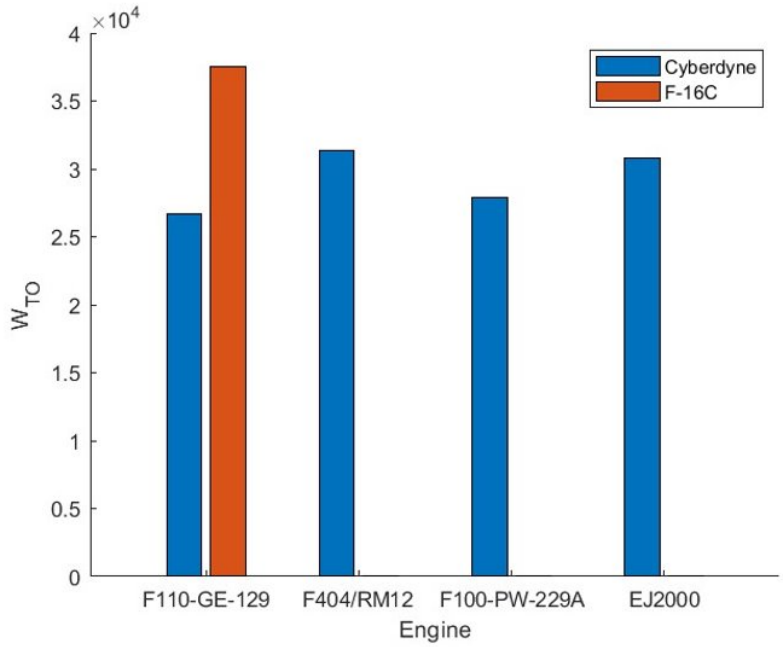


Figure 3.18: Engine Selection Trade Study

3.10.2 Engine Properties

Table 16 provided by General Electric shows the baseline performance of the F110-GE-129 engine [18]. The F110-129 provides a high thrust-to-weight ratio for the XQ-37 Reaver, solidifying the selection for better acceleration and rate of climb performance.

Table 16: Engine Performance Specifications (Sea Level/Standard Day)

F110-GE-129	English	SI
Thrust class	29,000 lb	129 kN
Length	181.9 in	4.6 m
Airflow	270 lb/sec	122.4 kg/sec
Maximum diameter	46.5 in	1.2 m
Bypass ratio	0.76	0.76

The XQ-37 has a thrust loading of 1.135 with a maximum afterburning thrust of 29474 pounds at sea level [3]. The SLS values for thrust and TSFC were obtained from the propulsion data in Nicolai Appendix J. The appropriate equations from Mattingly for a low bypass ratio turbofan engine with an afterburner were used to construct an engine deck for the F110-129 using the SLS values from Nicolai, as further performance characteristics are not publicly available for this engine [21]. Figures 3.19 and 3.20 show the maximum afterburning performance of the propulsion system from 0 feet to 50000 feet altitude.

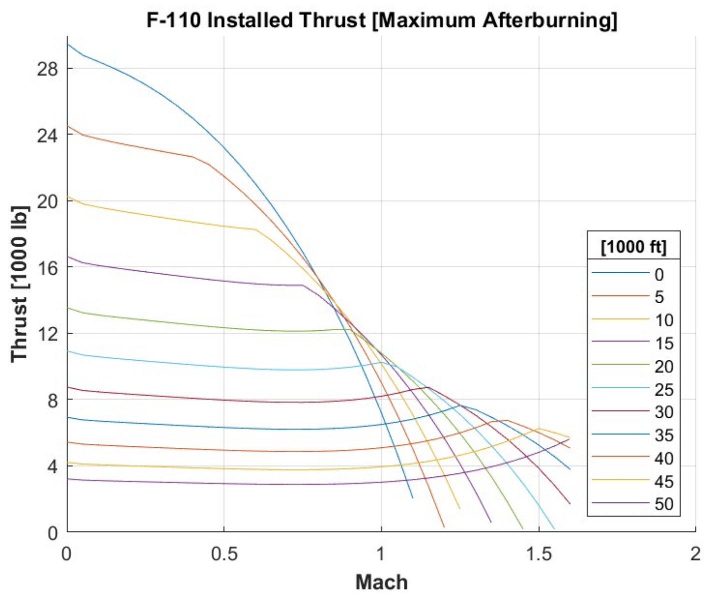


Figure 3.19: Maximum Afterburning Installed Thrust at Various Altitudes

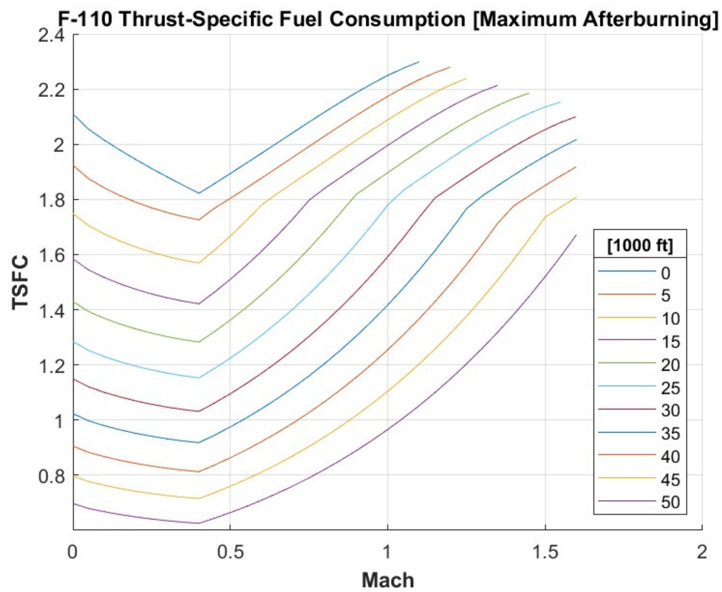


Figure 3.20: Maximum Afterburning TSFC at Various Altitude

3.10.3 Inlet Design

For the initial design of the inlet, a single underbody intake was considered. As the RFP requests ease of maintenance, it was decided that dual inlets were more suitable for the XQ-37 allowing simpler storage of the nose landing gear. This general inlet design decision not only simplified the front landing gear storage, but also created more space in the forward fuselage for avionics and fuel. Such progression of the inlet design can be visualized from Figures 3.21 and 3.22.

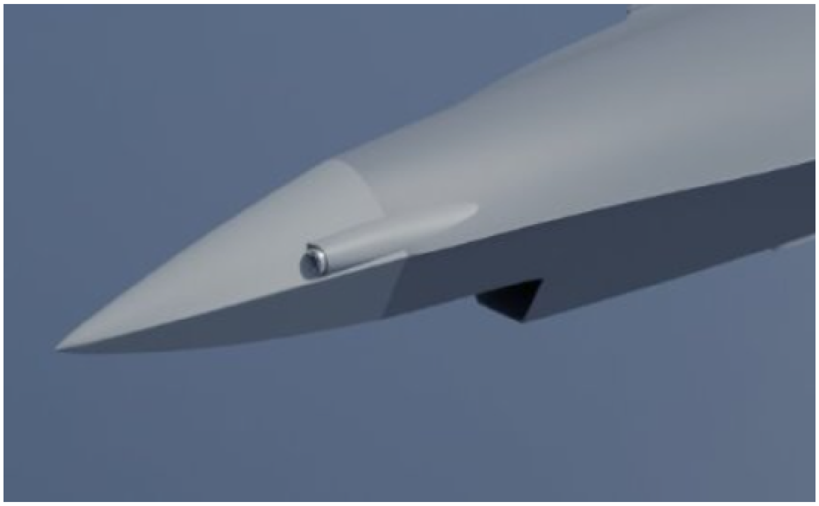


Figure 3.21: Initial Inlet Design (Single Underbody)

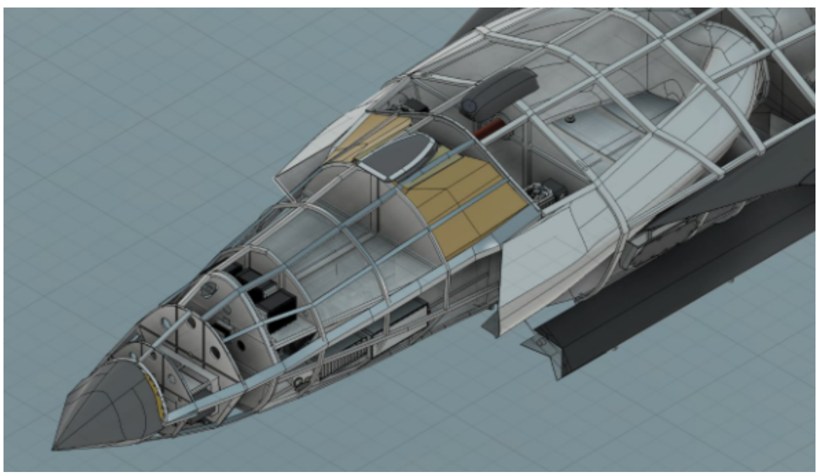


Figure 3.22: Dual Inlet Design on XQ-37 Reaver

As mentioned in Section 2.4.5, the inlet capture area was sized to be 3.81 feet squared per inlet for the dual inlet design. During the detailed design phase of the XQ-37, bypass doors will likely be needed to expel excess air during other mission segments. This design decision minimizes the amount of bypass drag for the total mission duration. While having a variable geometry inlet with bypass doors does add some complexity in the system, the diverterless supersonic intake and the fixed capture area reduce complexity significantly. Development of the DSI in the detailed design phase could also lead to efficiency up to Mach 1.6, eliminating the need for bypass doors and further simplifying the design.

3.11 Performance

3.11.1 Takeoff and Landing

Takeoff and landing performance was calculated using equations described in Nicolai which can be seen below [3].

$$S_{\text{TO}} = 20.9\frac{W/S}{\sigma C_{L\,\text{max}}(T/W)} + 69.6\sqrt{\frac{W/S}{\sigma C_{L\,\text{max}}}} \tag{6}$$

$$S_L = 79.4\frac{W/S}{\sigma C_{L\,\text{max}}} + \frac{50}{\tan\theta_{\text{app}}} \tag{7}$$

$$S_B = \frac{W_L}{g\mu\rho S_{\text{ref}}\left[\left(\frac{C_D}{\mu}\right) - C_{LG}\right]}\ln\left[1 + \frac{\rho S_{\text{ref}}}{2W_L}\left(\frac{C_D}{\mu} - C_{LG}\right)V_{\text{TD}}^2\right] \tag{8}$$

$$C_D = C_{D_0} + KC_{LG}^2 + \Delta C_{D_{\text{flaps}}} + \Delta C_{D_{\text{gear}}} + \Delta C_{D_{\text{misc}}} + \Delta C_{D_{\text{spoilers}}} \tag{9}$$

$$S_A = \frac{L}{D}\left[\frac{V_{50}^2 - V_{\text{TD}}^2}{2g} + 50\right] \tag{10}$$

Takeoff and landing distance estimates were calculated using Eq. (6) and Eq. (7) respectively. These empirical relations give us conventional takeoff and landing distances of 1780 ft and 6245 ft respectively. Since these were well within our 8000 ft requirement, more refined calculations were not performed. The greatest concern was icy runway landing conditions.

We chose a landing parachute with a drag coefficient of 1.2 for use in icy runway conditions. More refined calculations of landing distance involve three parts, airborne distance S_A, free roll distance S_{Fr}, and braking distance S_B. Airborne distance was calculated using Eq. (10) with a V_{50} of 307.8 ft/s, and a touchdown speed V_{TD} of 272.32. Both are a function of stall speed. This calculation resulted in an airborne distance of 953.057 ft. Assuming a reaction time of 3 seconds after touchdown, the freeroll distance was calculated to be 816.96 ft. Braking distance was calculated using Eq. (8) and Eq. (9). Landing drag buildup, including the parachute, results in a total drag coefficient of 1.85. With this drag, an icy μ of 0.07, and a full weight aircraft, the braking distance is 2057 ft. Totaling these distances we achieve an icy runway landing distance of 3828 ft. The icy runway landing distance is shorter than the conventional landing distance because of the significant drag induced by the parachute, with a drag coefficient of 1.2.

Table 17: Takeoff and Landing Parameters

Parameter	Value
TOGW	25,973 lbs
Thrust	29833 lbs
S	377.74 ft^2
Takeoff Distance	1780 ft
Conventional Landing Distance	6245 ft
Icy Runway Landing Distance	3828 ft
Takeoff CL with High Lift Devices	1.31
Stall Speed	236.8 kt
Takeoff Speed	182.39 kt

3.11.2 Range and Endurance

The payload-range diagram as seen in Figure 3.23 was created using Bruget's range equation for ferry which is described by Nicolai [3]. This considers an aircraft fuel capacity of 9100 lbs and a maximum takeoff weight of 25973 lbs. Our missions require 1300 lbs of payload. A 5500 lbs payload is achievable by trading fuel weight. The diagram was created with optimal performance in consideration. A higher range with 5,500 lbs of payload may be possible with reduced performance metrics. This range can be increased further by using extra fuel tank space, but reduced performance metrics would have to be accepted.

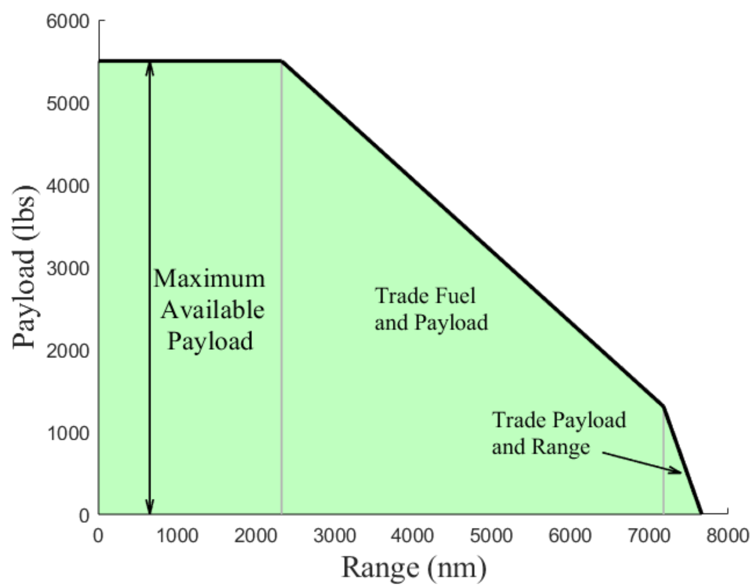


Figure 3.23: Payload Range Diagram

The Mattingly equations [21] were used to determine TSFC at varying flight conditions, and lift over drag was estimated using analytical equations from Nicolai [3] to determine range and endurance factors. Plots of this analysis can be seen in Figure 3.24 and Figure 3.25. As a result of these analyses an optimal range and endurance speed were determined. Endurance at 35,000 feet has an optimal speed of Mach 0.7, and range is optimized at 45,000 ft with a speed of Mach 0.775.

Refined lift over drag values described in section 3.7.4 are used to calculate mission performance.

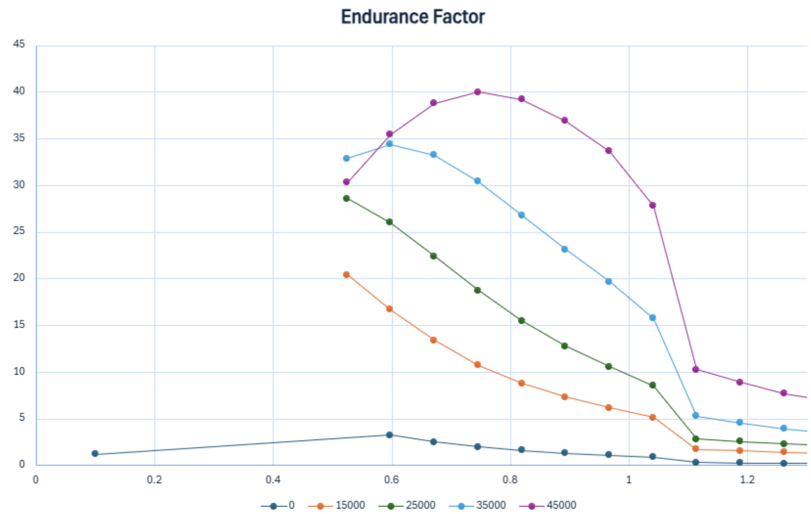


Figure 3.24: Endurance Factor at Different Altitudes

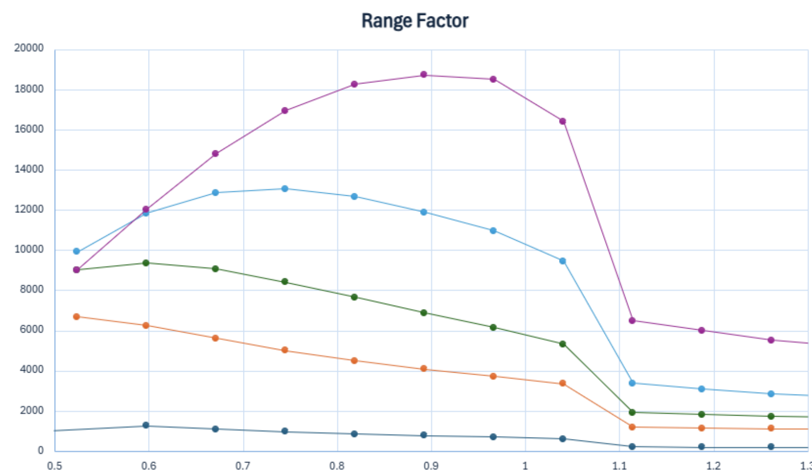


Figure 3.25: Range Factor at Different Altitudes

3.11.3 Specific Excess Power Contours

Figure 3.26 shows the Specific Power contours calculated with a load factor of one, and 50 percent internal fuel stores with full payload. It is done throughout the allowable Mach range from M = 0.1 to the max Mach of M = 1.6. For calculating the specific power, thrust was found using the Mattingly equations [21] along with the engine characteristics of the F110-GE-129 at sea level given by the manufacturer. Drag elements were calculated via the analytical equations from Nicolai [3].

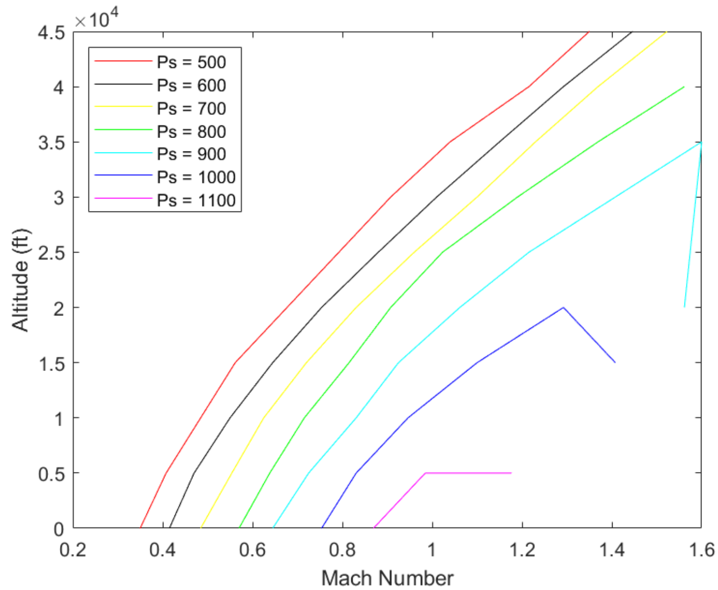


Figure 3.26: Specific Excess Power Contours at Varying Altitudes with a Load Factor of 1

3.11.4 Time to Climb

The time to climb was calculated using the specific excess power relationship from Nicolai [3]. A climb schedule was generated at ranges of altitude ranging from 0 to 55,000 feet above sea level. In order to fulfill the RFP requirement of reaching an altitude of 35,000 ft over a distance of 4.8 nautical miles in under 60 seconds, the climb schedule with maximum climb angle was used, shown in Figure 3.27.

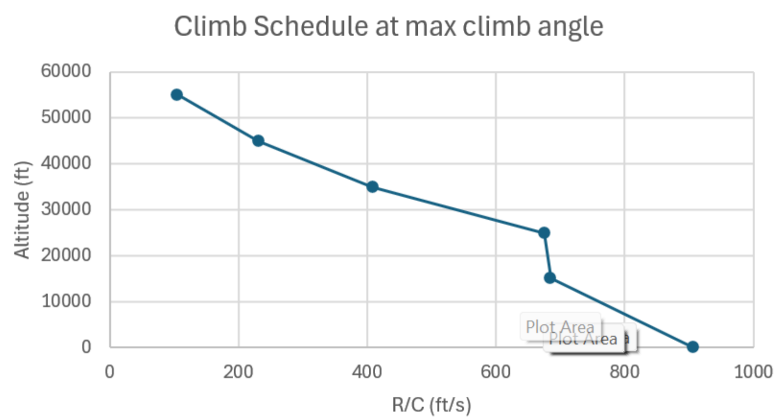


Figure 3.27: Climb Schedule at Maximum Climb Angle After Takeoff

In order to find the time to climb, the climb schedule was integrated over partial segments, then summed over the total distance. The Climb schedule for maximum rate of climb, shown in Figure 3.28, is able to reach 35,000 feet in 47 seconds, but over a distance of 5.65 nautical miles.

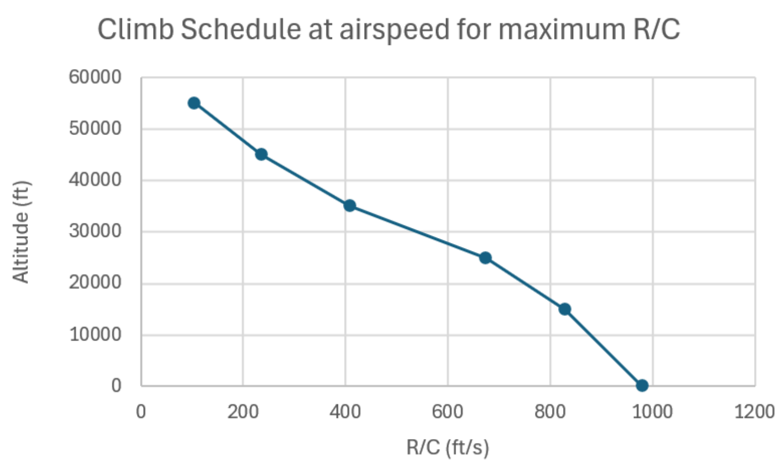


Figure 3.28: Climb Schedule at Maximum Rate of Climb After Takeoff

3.11.5 V-Speeds

V-Speeds were taken from separate sub-teams and compiled in Table 18.

V-Speed	Name	Indicated Airspeed (IAS)
V_S	Stall Speed (clean)	284.5 kt
V_{S0}	Stall Speed (landing config)	236.8 kt
V_X	Best Angle of Climb	546.7 kt
V_Y	Best Rate of Climb	688.4 kt
V_A	Maneuvering Speed	246.5 kt
V_R	Rotation Speed	307.8 kt
V_{GLIDE}	Best Glide Speed	400.6 kt
V_D	Dive Speed	701 kt

Table 18: V-speeds

3.12 Structures and Materials

3.12.1 Material Selection

A critical phase in the aircraft design process involves the selection of appropriate materials for all structural component. These material selections are directly impacted by structural integrity, overall performance, and manufacturing feasibility. Material specifications for each component were established through consideration of anticipated operational loads, environmental factors, and weight constraints. Table 19 provides a preliminary summary of the selected materials for major structural components, outlining their designated location and the technical justifications for each choice.

In evaluating suitable materials for the aircraft structure, aluminum alloys and composite materials emerged as the primary candidates due to their favorable strength-to-weight characteristics.

Table 19: Material Selection and Justification

Structural Component	Location	Material Selection	Justification	Density (lb/in^3)
Spars	Wing Box	7075-T6 Aluminum	good strength and fatigue resistance at low cost	0.102
Skin	Wing	Graphite/Epoxy Composite (IM7/8552)	lightweight, high strength, fatigue resistance	0.057
Ribs	Wing Box	2024-T3 Aluminum	good strength-weight ratio at low cost	0.1
Frames	Fuselage	2024-T3 Aluminum	good strength-weight ratio at low cost	0.1
Bulkheads	Wing Root/Mid Fuselage	Titanium (Ti-6Al-4V - Grade 5)	high temperature resistance, strong	0.16
Bulkheads	All Other Bulkheads	7075-T6 Aluminum	good strength and fatigue resistance at low cost	0.102
Longerons	Nose Region	Pultruded Carbon Fiber	very light, not a huge lad bearing area	0.0577
Longerons	Mid Fuselage	7075-T6 Aluminum	high strength, affordable, not experiencing high heats	0.102
Longerons	Wing Root and Around Engine	Titanium (Ti-6Al-4V - Grade 5)	high temperature resistance, strong	0.16
Skin	Fuselage	Graphite/Epoxy Composite (IM7/8552)	lightweight, high strength, fatigue resistance	0.057
Engine Structure	Engine	Titanium (Ti-6Al-4V - Grade 5)	high temperature resistance, strong	0.16
V-Tail Skin	Empennage	Graphite/Epoxy Composite (IM7/8552)	lightweight, high strength, fatigue resistance	0.057
V-Tail Spars	Empennage	7075-T6 Aluminum	good strength and fatigue resistance at low cost	0.102
V-Tail Ribs	Empennage	T300/epoxy	lightweight, moderately strong	0.0056
Ribs - Flaperon	Wing	IM6 Carbon Fiber	reduces weight and maintaining stiffness	0.057
Spars - Flaperon	Wing	IM6 Carbon Fiber	reduces weight and maintaining stiffness	0.057
Skin - Flaperon	Wing	Graphite/Epoxy Composite (IM7/8552)	lightweight, high strength, fatigue resistance	0.057
Ribs - Aileron	Wing	IM6 Carbon Fiber	reduces weight and maintaining stiffness	0.057
Spars - Aileron	Wing	IM6 Carbon Fiber	reduces weight and maintaining stiffness	0.057
Skin - Aileron	Wing	Graphite/Epoxy Composite (IM7/8552)	lightweight, high strength, fatigue resistance	0.057
Doors	Weapons Bay	Graphite/Epoxy Composite (IM7/8552)	lightweight, high strength, fatigue resistance	0.057
Hinges	Weapons Bay	Titanium (Ti-6Al-4V - Grade 5)	high temperature resistance, strong	0.16

These materials were chosen for their well-established structural performance, cost-effectiveness, and compatibility with aerospace manufacturing methods. A more detailed list of their physical properties can be seen below in Table 20. These values are used in the stress and buckling calculations, which will be described in detail in a later section. Note that the values for the Graphite Epoxy composite are considering a 0° (unidirectional) layup.

Table 20: Material Properties [1][2]

Material	Density (lb/in^3)	Yield Strength (psi)	Ultimate (Tensile) Strength (psi)	Young's Modulus (ksi)	Material Cost
7075-T6 Aluminum	0.102	73000	83000	10400	low
Graphite/Epoxy Composite (IM7/8552)	0.057	-	373000 - 395000	23800	mid - high
2024-T3 Aluminum	0.1	50000	70000	10600	low

3.12.2 V-n Diagram

To determine the aircraft loading at various airspeeds, a V-n diagram was developed. Figure 3.29 depicts the XQ-37 Reaver's operations flight envelope at 35,000 ft Mean Seal Level (MSL). The V-n diagram shows the relationship between load factor (n) and equivalent airspeed (EAS).

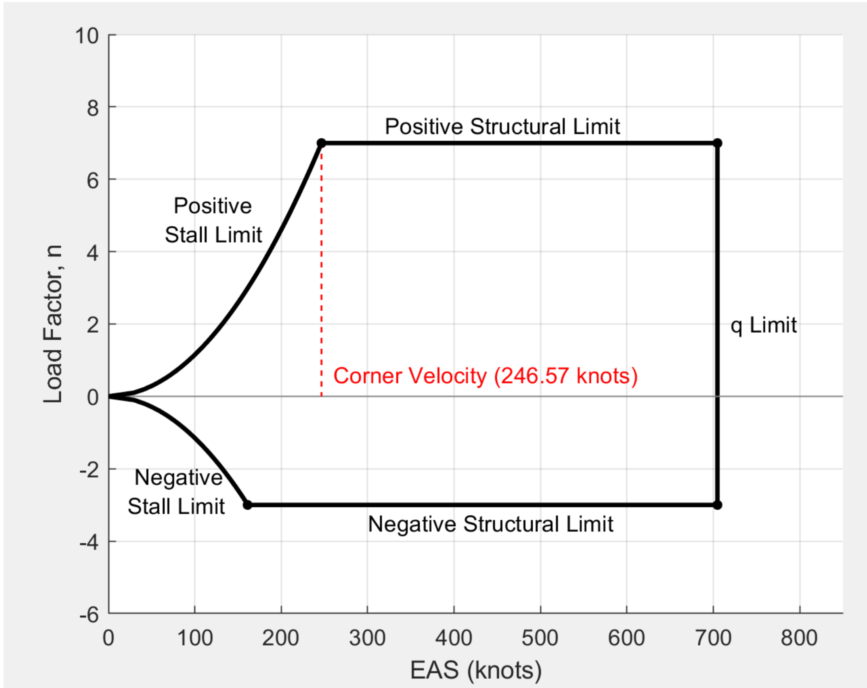


Figure 3.29: Vn Diagram at 35,000 ft MSL

As can be seen above, this flight envelope is bounded by aerodynamic and structural limits. The curves on the left hand side of the diagram represent the positive and negative stall limits, showing the minimum speeds required to achieve various load factors before the wing stalls. The horizontal lines at the top and bottom of the diagram represent the positive and negative structural limits, representing the maximum load factors the aircraft is capable of withstanding. Finally, the vertical line on the far right side of the diagram marks the dynamic pressure limit (q limit), which establishes the maximum allowable airspeed. It is also important to know that the corner velocity of this aircraft is approximately 246.57 knots. This represents the lowest speed at which the maximum positive load factor can be reached.

3.12.3 Wing Box Analysis

As mention above, a series of stress and buckling analyses were completed in order to determine the front and rear spar locations at the root of the wing, as well the skin thickness, spar thicknesses, and rib spacing within the wing box. To verify that the preliminary wing box will not yield or buckling during flight, the stress in the spars and skin were calculated at the four critical flight conditions determined by the aircraft's flight envelope. These conditions are represented as the four "corners" in the V-n diagram.

Several assumptions were made in order to calculate the stress values at the root of the wing. The first is that these calculations assume thin walled approximations. The wing box is also approximated as a rectanglular box which results in an assumption of the front and rear spars being the same height. A constant skin thickness and spar thickness was assumed, as well as a constant rib spacing. Finally, no taper was taken into account. All conditions are using an approximated, elliptical lift distribution along the span of the wing. Finally, as required by the RFP, a factor of safety of 1.5 was used for applicable calculations. All of these assumptions lead to a more conservative estimate of the stresses involved in the wing box. A more detailed and accurate analysis can be done in the detailed design phase using FEA software.

The stress values and plots shown below correspond to the condition of maximum positive load factor at zero angle of attack, represented by the top-right corner of the V-n diagram. This flight condition imposes the highest structural demand on the wing box and is therefore the most critical for structural evaluation. While all four corner conditions were analyzed to ensure structural integrity across the flight envelope, only the most demanding case is presented here to conserve space in the report. Table 21, below, gives the stress values of the spars and skin at the wing root for this critical flight condition. Note that the shear stress in the spars includes the transverse and torsional shear stress, while the skin only experiences a torsional shear stress. As can be seen, these stress values are less than the yield stress values for the respective material properties. This means that the wing box will not yield during critical flight condition, ensuring a safe design.

Table 21: Stress Values at the Wing Root

	Spars	Skin
Shear Stress (psi)	2339.4	24953.6
Bending Stress (psi)	32060.1	16030.1
Von Mises Stress (psi)	32098	29657.6

Figure 3.30, below, displays the approximated lift curve along the length of the wing, as well as the shear force and moment diagrams. Note that the wing was approximated as a cantilever beam, so the shear force is zero at the root, and moment is zero at the wing tip. Figure 3.31, below, displays the approximated bending stress and torsional shear stress of the wing skin along the span of the wing. Finally, Figure 3.32 displays the bending stress, torsional shear stress, and transverse shear stresses along the span of the wing.

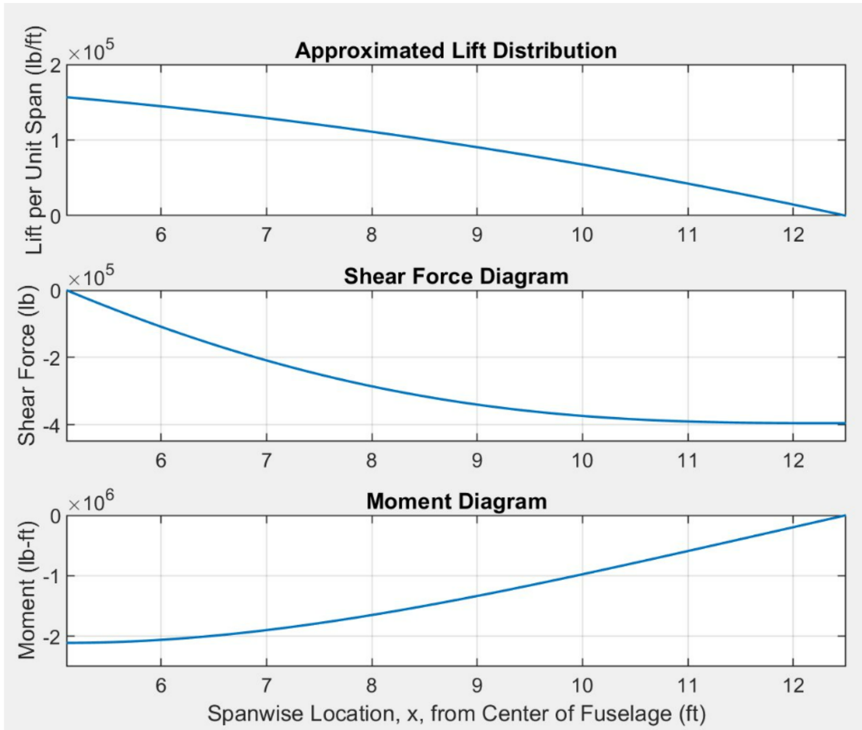


Figure 3.30: Wing Box Lift Distribution, Shear Force, and Moment Diagram

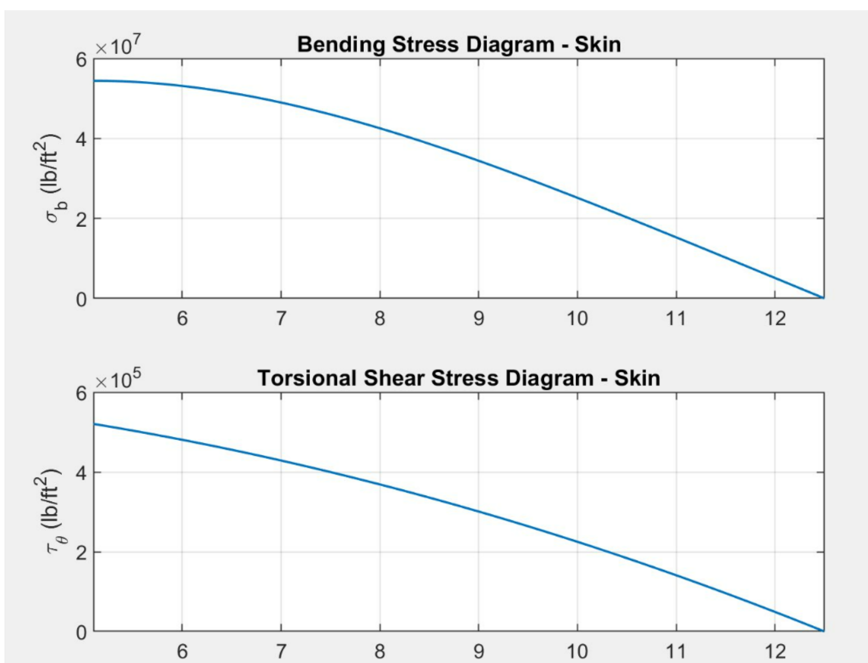


Figure 3.31: Wing Box Skin Stress Plots

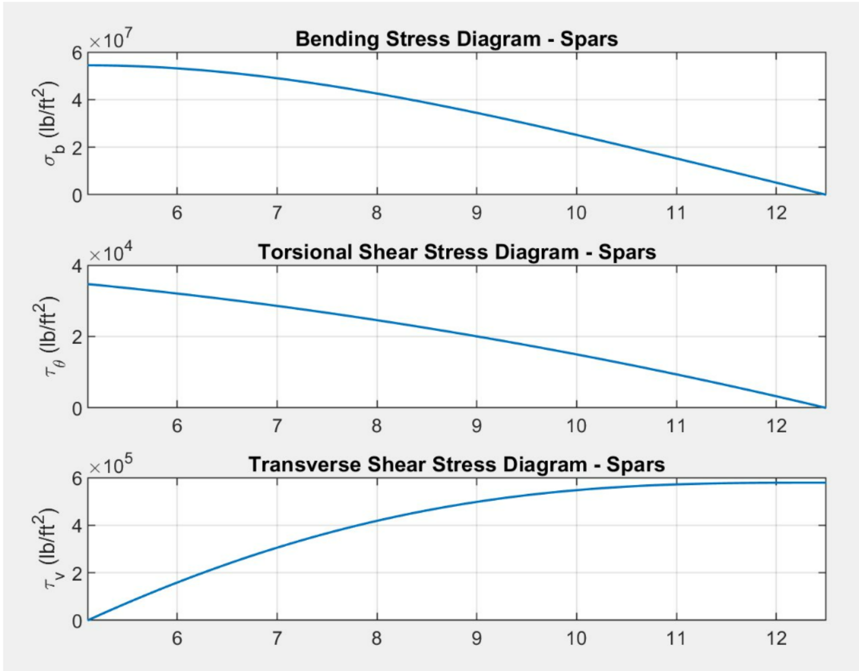


Figure 3.32: Wing Box Spar Stresses

Table 22, below, presents the critical buckling coefficients and corresponding stress values for both the wing box skin and spars. As shown, these values are greater than the calculated shear stresses listed in Table 20, indicating that the wing box structure remains stable and will not experience buckling under the critical loading condition.

Table 22: Critical Buckling Stresses in the Wing Box

	Buckling Coefficient	Critical Buckling Stress (psi)
Skin (shear)	5.366	35700
Spars (shear)	135.519	31300
Compressive (normal)	154.302	1240

3.12.4 Wing Box Design

The finalized wing box layout defines the core internal structure of the wing, made up of five longitudinal spars and ten transverse ribs, all enclosed by an outer skin with a uniform thickness of 0.05 inches. This structure is designed as a wet wing, meaning the internal volume of the wing box itself is sealed and used to store fuel. Together, the left and right wings can hold up to 1400 pounds of fuel.

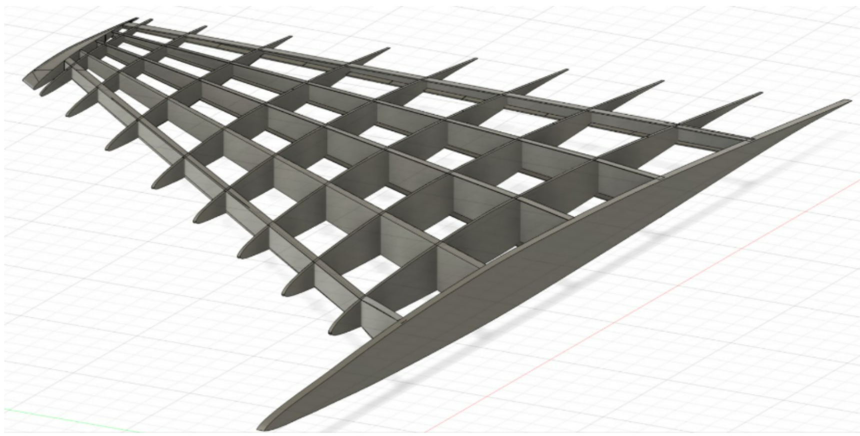


Figure 3.33: Final Wing Box Design

The wing box has five C-channel spars that handle the bulk of the bending and shear load. Each spar is 0.8 inches thick at the root and tapers across the wingspan with a ratio of 0.3125. The top and bottom flanges of the C-channel are 0.3 inches thick and 0.17 inches wide. The front spar is located where the leading edge slats of the wing hinge. This ends up being about 6.85% of the root chord. The rear spar sits at about 80.39% of the root chord, supporting the trailing edge control surfaces (ailerons and flaps). An up close depiction of one of the spars can be seen below in Figure 3.34.

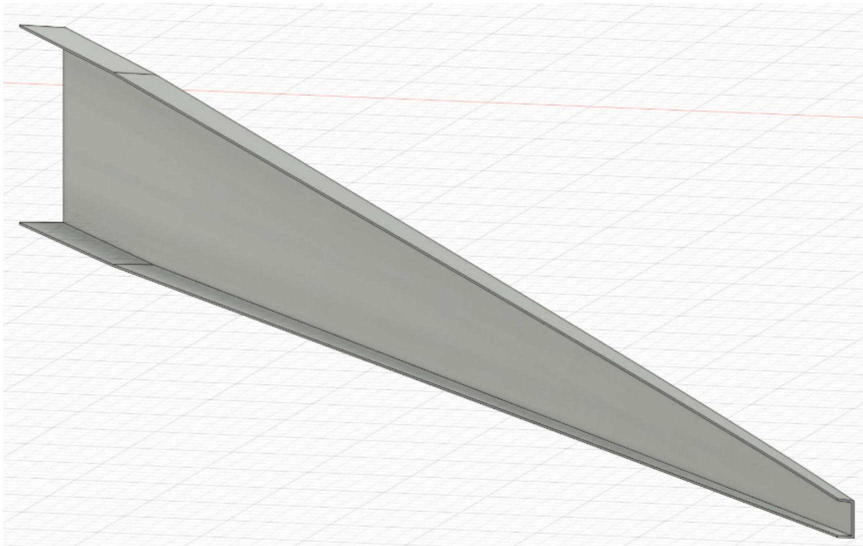


Figure 3.34: Wing Box Spar Design

The wing's internal rib structure plays a critical role in maintaining the airfoil profile and distributing loads throughout the wing box. The final wing box design consists of ten ribs, running spanwise across the wing. The are evenly spaced at 10 inch intervals, and have a web thickness of 0.1 inches. Each rib includes a top and bottom flange mimicking a C-channel profile, just like the spars. These flanges are both 0.1 inches thick and 0.6 inches wide, adding stiffness and helping the ribs resist local buckling. Initial rib placement was determined based on the locations of the control surfaces, with ribs positioned at the start and end of each surface. The remaining ribs were then added between these points and evenly spaced along the span. Figure 3.35, below, shows an up close view of the final rib design.

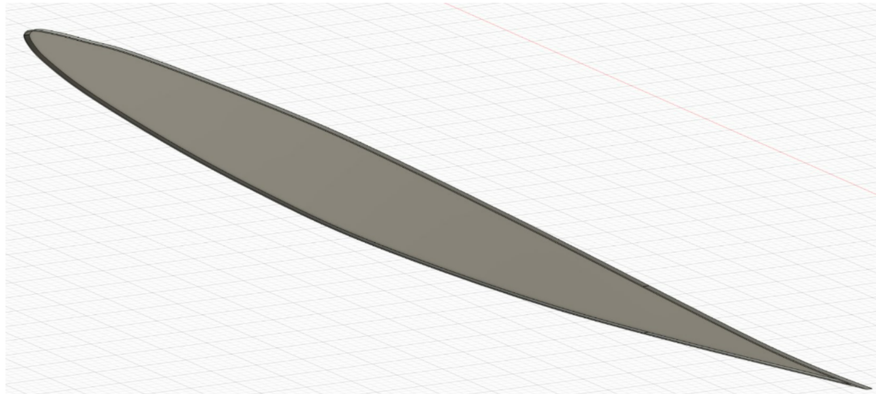


Figure 3.35: Wing Box Rib Design

The final internal wing structure can once again be seen in Figure 3.36, below. This side-by-side view of the wing and its internal structure provides a clearer picture of how the control surfaces align with the front and rear spars. It also highlights how the rib locations correspond to the start and end points of the control surfaces—particularly in the region between the flaps and ailerons.

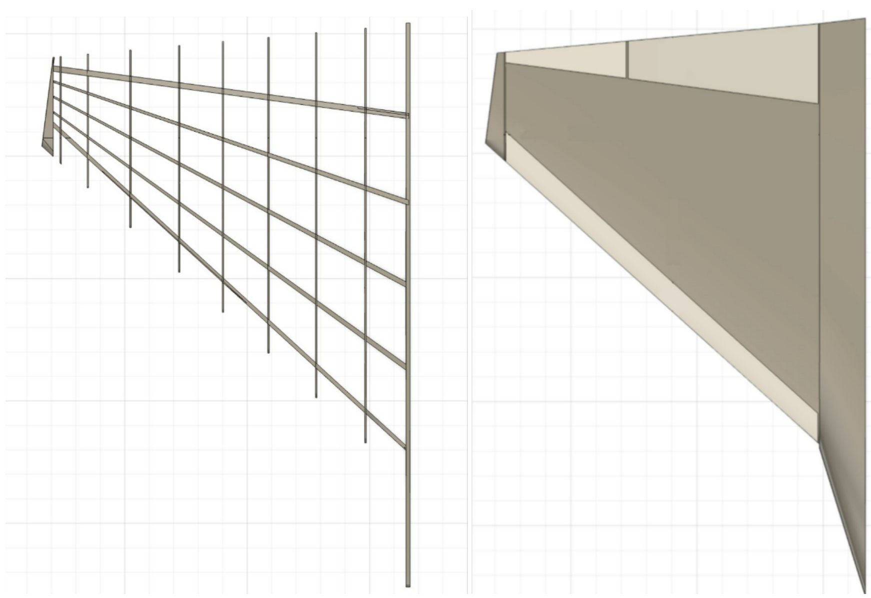


Figure 3.36: Side-By-Side Top view of Wing and Internal Structure

3.13 Subsystems

3.13.1 Landing Gear

The landing gear configuration was designed based off the USAF requirements for controllability and balance. These standards state that the tip back angle of the aircraft must be between 16 and 25 degrees, to prevent a tail strike during takeoff or landing. The overturn angle must be less than 63 degrees, and the nose gear must hold between 5 and 20% of the aircraft to maintain steering capabilities and balance during taxi.

The XQ-37 has a tip back angle of 18 degrees, as shown in Figure 3.37 below. The weight distribution leaves 18% of the aircraft weight held by the nose gear.

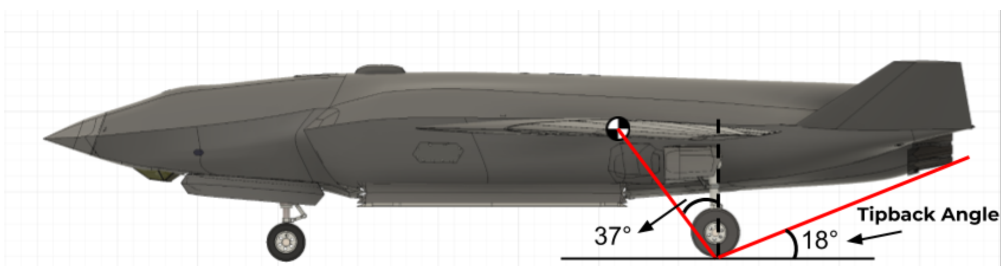


Figure 3.37: Tipback Angle

The overturn angle was calculated using Equation 11 below. This equation uses the height of the center of gravity, h_{cg}, which is 6.022 feet. It also uses the distance between the nose gear and center of gravity, l_n , and the δ angle. Both of these measurements are seen in Figure 3.38 below. Together these numbers give the XQ-37 an overturn angle of 58.27 degrees.

$$\psi = \arctan(\frac{h_{cg}}{l_n \sin \delta}) \tag{11}$$

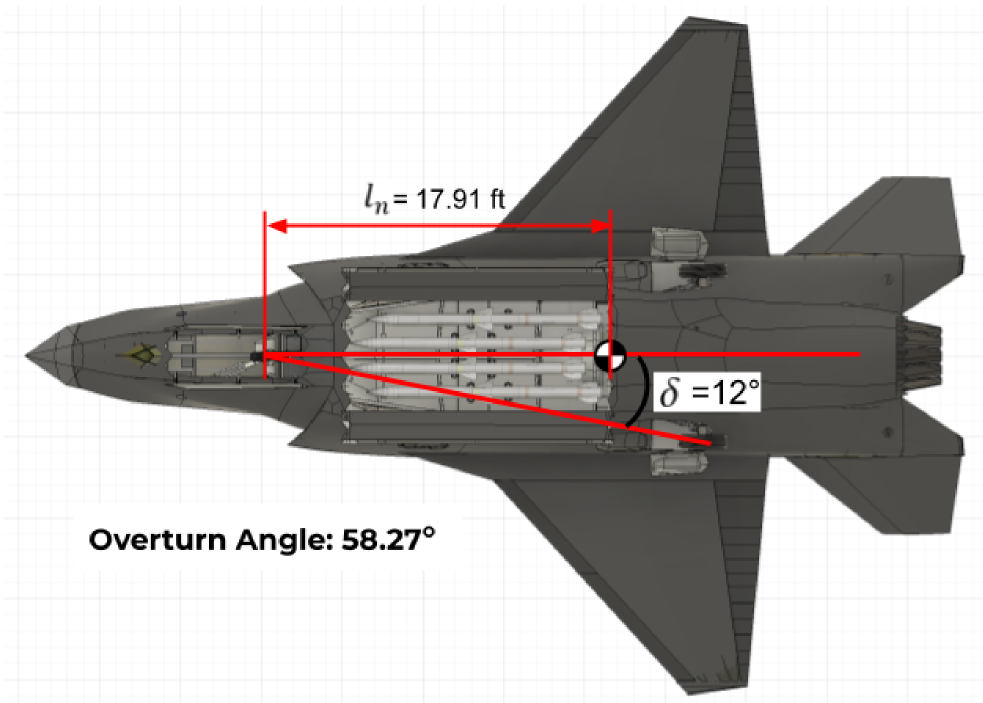


Figure 3.38: Overturn Angle

3.13.2 Vehicle Systems Architecture

The XQ-37 requires many advanced interconnected subsystems to accomplish its missions. In order to verify these systems were viable, an aircraft architecture diagram was created and can be seen in Figure 3.39 below. It encompasses all of the main subsystems present in the aircraft and ensures redundancy is in place throughout the mission critical systems.

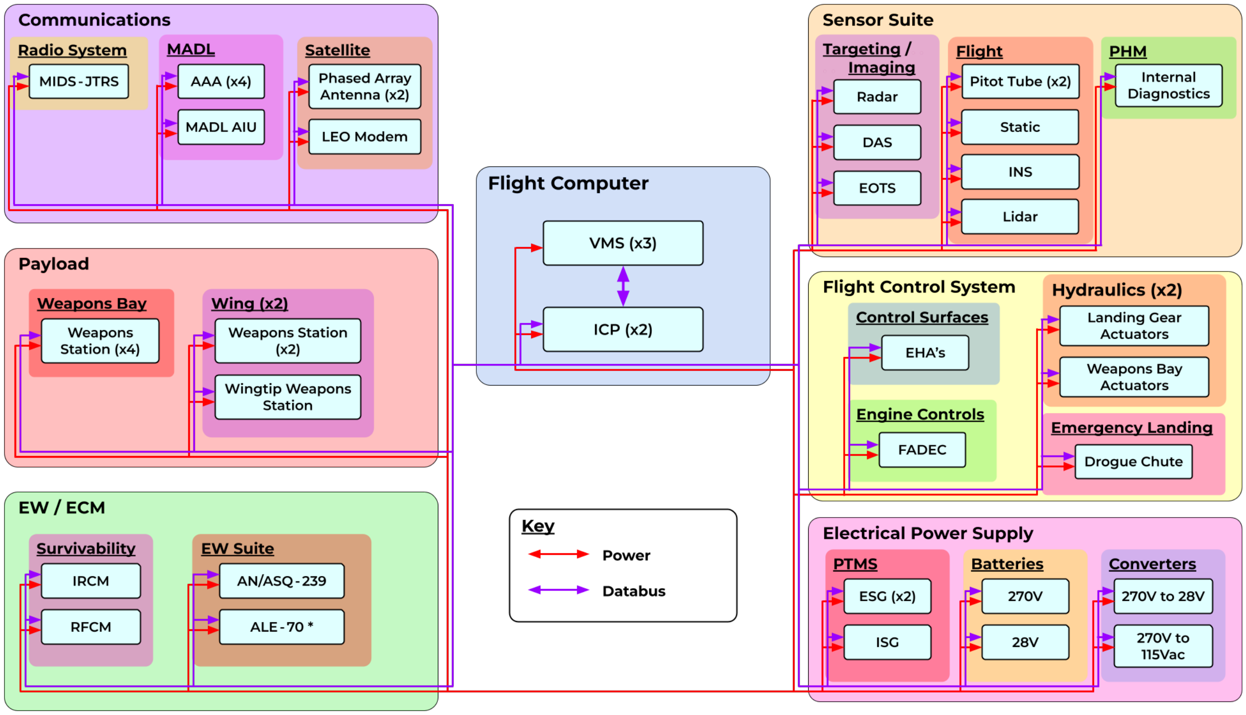


Figure 3.39: Aircraft Architecture Diagram

3.14 Project Management

3.14.1 Cost Breakdown

Given the stringent cost constraints put forth by the RFP, paired with the sensitivity of design decisions on overall pricing, Cyberdyne Aerospace decided to develop a comprehensive cost model to evaluate trade-offs between cost and utility in the most accurate way possible at this stage of our design process. To do this, our team utilized a modular cost analysis function based on the model described in Jan Roskam's Airplane Design, Part VIII: Airplane Cost Estimation. Based on this, a MATLAB script was designed to dynamically update our team's cost estimation as design changes were made. This verified compliance with the RFP's 25 million dollar unit cost constraint. The results of this cost estimation model are presented in the tables below.

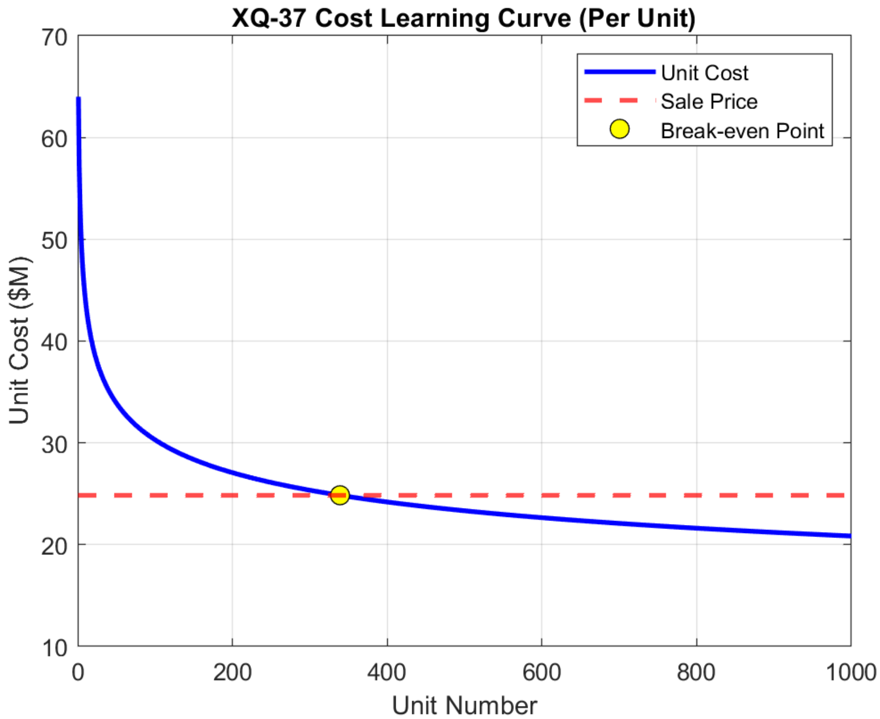


Figure 3.40: XQ-37 Reaver Cost Learning Curve

Table 23: Detailed Cost Breakdown for the XQ-37 Reaver

Parameter	Value
Flyaway Cost	$24.68M
R&D Cost	$1.38M
Manufacturing Cost	$21.34M
Acquisition Cost	$23.47M
Procurement Cost	$29.85M
Operational Cost	$59.91M
Disposal Cost	$0.85M
Total Lifecycle Cost	$85.61M
1st Unit Cost	$64.5M
Learning Curve	90%
Break-even Point	339 Units

There are a few notable aspects of this cost analysis that are worth mentioning. First and foremost, it is important to note that the two main drivers of price in this model are empty weight, and top speed. These two variables effect almost every aspect of the design process, and are directly proportional in an increase of price for the aircraft. Therefore, it was imperative to push these numbers down as much as possible in order to meet our cost requirements. Secondly, Although the software costs are a significant portion of the overall cost of the aircraft, the method outlined by Roskam in this version did not have a direct calculation, and was instead built into both the research and development, and manufacturing costs of the aircraft. Third, for the learning curve shown in Figure 3.40, many of these values associated with the plot, such as the learning curve rate, break-even point, and initial unit cost are estimates suggested by the methodology put forth in Roskam's book, and are by no means exact values. Lastly, as comprehensive as this cost model may seem, it is still an estimate for the overall price of the aircraft, and cannot measure a multitude of variables that will influence the overall aircraft price down the line. Therefore, it is important to continue to refine and upgrade this cost model to more accurately reflect our unit cost as this program moves forward.

3.14.2 Technical Risk

The unmanned nature of the aircraft naturally presents the issue of cybersecurity risk for the aircraft, which inherently threatens its operational integrity. These risks pose a monumental threat to public safety and disrupt the intended functionality of the systems, as a single cybersecurity breach could render the aircraft useless. Addressing cybersecurity is crucial to maintaining trust and ensuring reliable performance in unmanned operations within the national domain. Potential security issues include signal jamming, GPS spoofing, malware, data breaches, and more, with consequences ranging from compromised functionality and financial losses to safety concerns and national security risks. Before mitigations, this technical risk was assessed as very high, placing it in the red zone of the risk matrix, as shown in Figure 3.41. This score is determined by assessing the likelihood of the risk occurring and its potential consequences. Ultimately, the high probability of attacks, fueled by automation and growing digital vulnerabilities add to the potentially severe consequences, raising the score.

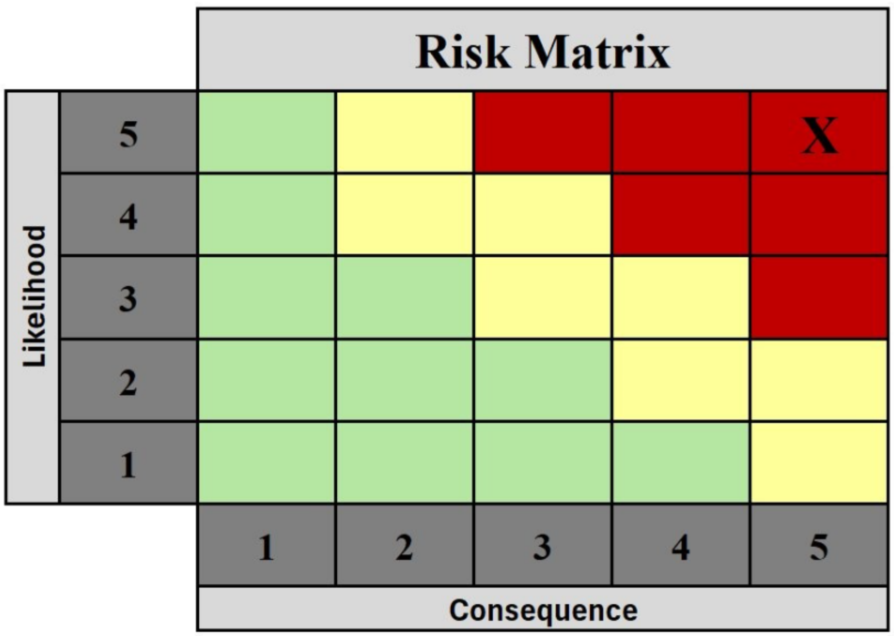


Figure 3.41: Cybersecurity Risk Matrix (before mitigations)

To mitigate this risk, several measures were implemented, including high-level military-grade encryption for communication channels further outlined in the proposal section of this document, and the use of a vulnerability monitoring system. System-level penetration testing will also be conducted during both the design and manufacturing stages, supported by a secure software development program encompassing the entire program, including maintenance systems used in and around the aircraft. Finally, additional measures will include establishing an air-gapped engineering environment, implementing strict user access controls, and hardening system defenses wherever possible. These mitigations lowered the risk level to the yellow zone, indicating a more secure environment, though still not negligible. This matrix can be seen in Figure 3.42. Given the critical threat this risk poses to the mission and to public safety, it will remain ever-present. As a result, the mitigations will adapt as technology advances.

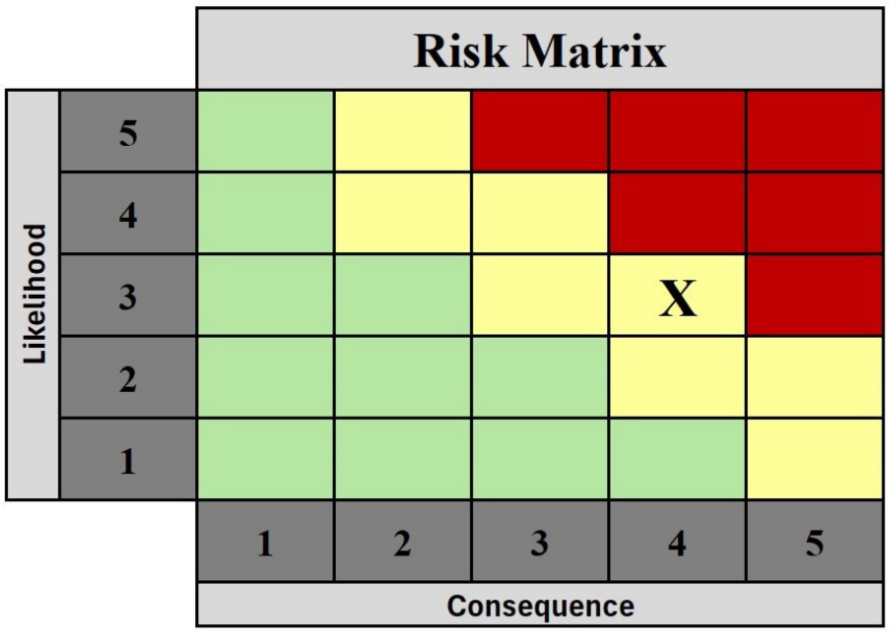


Figure 3.42: Cybersecurity Risk Matrix (after mitigations)

Given that the aircraft is designed to be near marginally stable to maximize maneuverability and that there is inherent uncertainty in the SM and CG calculations, the aircraft is at risk of being longitudinally unstable. Additionally, as discussed in the stability and control section, for non-optimized fuel distribution the aircraft has a negative SM meaning it is unstable in that case. Therefore, without proper precautions, there is a risk that the aircraft could become unstable and the pilot could lose control of the aircraft which could pose a threat to the public and result in mission failure. Given the several potential sources for instability and the dire consequences of an unstable drone fighter, the risk level is in the red zone as shown in Figure 3.43.

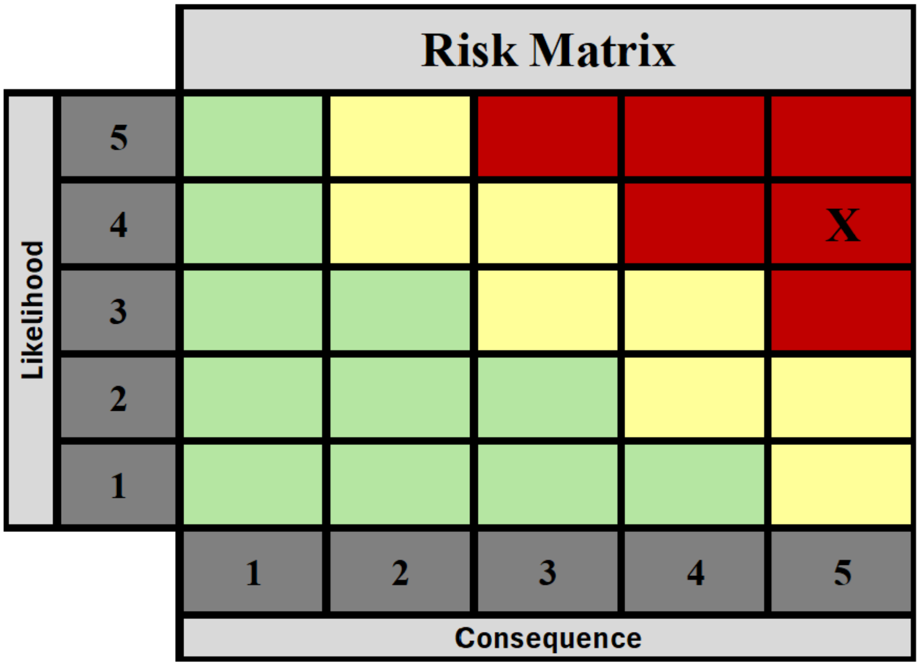


Figure 3.43: Aircraft Stability Risk Matrix (before mitigations)

Several mitigations are in place to reduce this risk. First, a triple-redundant digital flight control system is employed to ensure the aircraft remains stable and controllable by the pilot. As an additional benefit this faciliates the implementation of a return to home feature in the case of communication failure. Second, the aircraft features an optimized fuel distribution system to minimize CG travel and gives the aircraft a positive SM. Third, stability anaylsis was completed over the non-optimized CG travel range to ensure the aircraft meets the SM requirement even in this case. Given the multiple mitigation strategies the likelihood of the risk decreases. Additionally, the consequence of the risk decreases since multiple systems are in place to address the risk, and so there is less of a consequence if one system is to fail. The risk level is therefore in the green zone as seen in Figure 3.44 following the before mentioned mitigations.

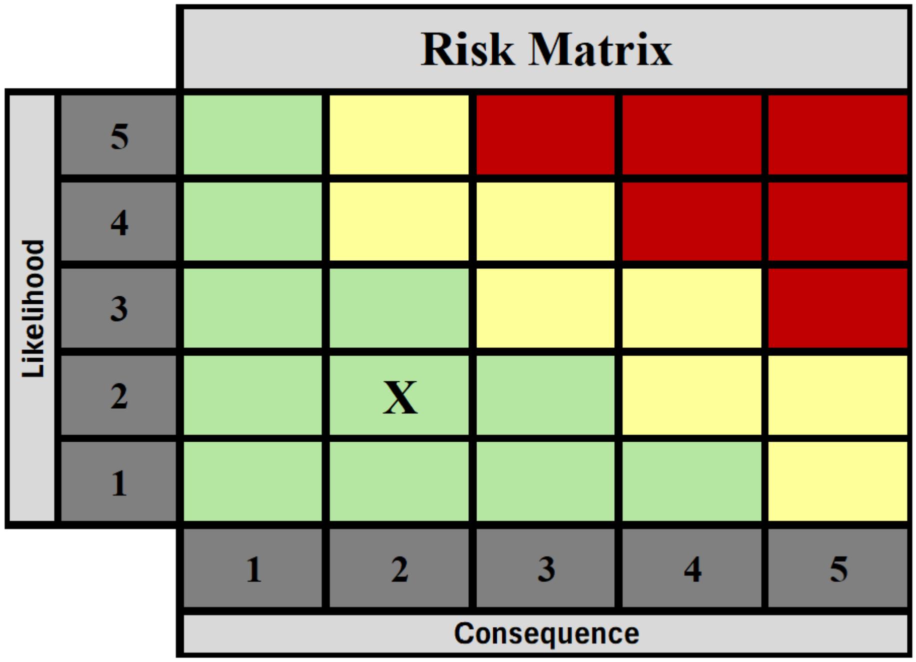


Figure 3.44: Aircraft Stability Risk Matrix (after mitigations)

3.14.3 Program Risk

Given the complexity and advanced capabilities required for designing the HDI, there is a significant risk that the unit cost may exceed the budget of 25 million [2025 USD], potentially leading to compromises in performance, technology, or project scope. This risk was initially assessed as being high, reflected by its placement in the red zone of the risk matrix, seen in Figure 3.45.

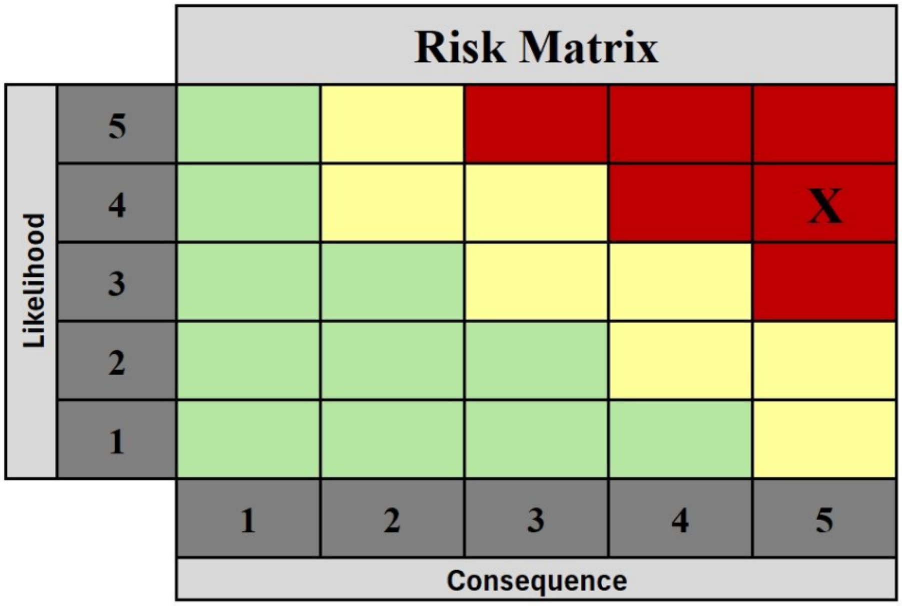


Figure 3.45: Program Risk Matrix (before mitigations)

This designation reflects the potential for budget overruns and the severe consequences for two scenarios: the design not being realized due to excessive costs, and/or cost reductions leading to performance compromises. To mitigate this risk, a fully unmanned aircraft was designed to eliminate the complexities and higher costs associated with manned capabilities. Furthermore, the size, weight, and top speed of the design were restricted, directly limiting the unit cost. Additionally, a comprehensive cost model was established using the methodology outlined in *Jan Roskam's Airplane Design, Part VIII: Airplane Cost Estimation* [22]. This allowed the team to consistently update design decisions in conjunction with cost projections to determine the practicality of major design changes. Strong project management practices were used to ensure adherence to the strict schedule depicted in the project plan and EIS timeline, as seen in the sections below. These mitigations ultimately lowered the risk to being in the yellow zone, seen in Figure 3.46. Due to the high complexity of designing the HDI, as well as any potential unforeseen complications, a risk of exceeding budget will always exist. This risk, however, is accompanied by enhanced performance and the technological capabilities needed for mission success.

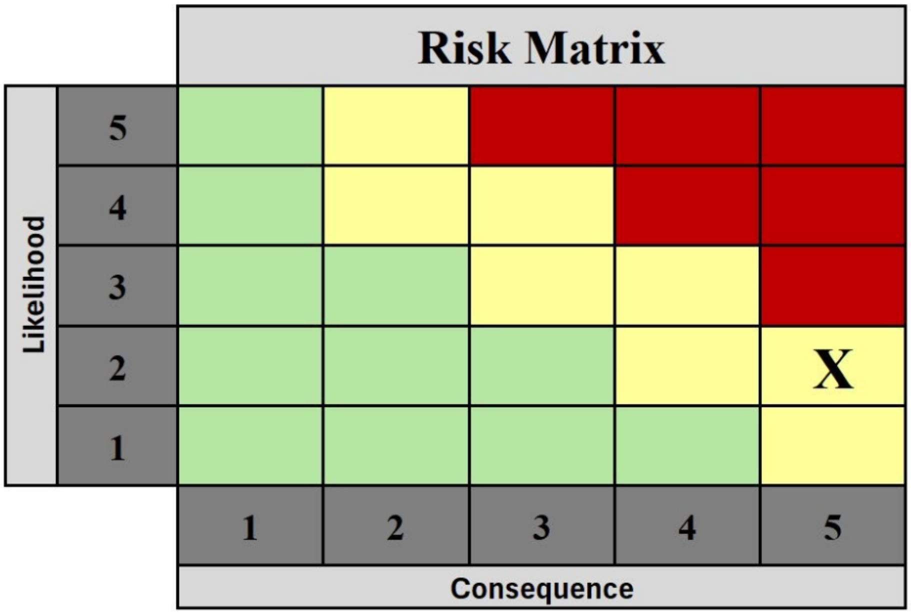


Figure 3.46: Program Risk Matrix (after mitigations)

3.14.4 EIS Timeline

To provide a clear path for future development of this platform, it is necessary to define a detailed plan spanning from conceptual design to entry into service. As such, Cyberdyne Aerospace has created an Entry Into Service timeline, shown in Figure 3.47. As can be seen, this plan consists of three phases: Design, Testing, and Manufacturing. The design phase contains all tasks relating to analytic calculation and verification, making sure the vehicle will perform the mission to the highest of merit under the constraints given, as well as considering new technologies and other abilities that we think can unlock more capability for the vehicle. Our team will also model the aircraft in its entirety, and sort out prototyping logistics to transition to the next phases with ease. After the design phase, we move into the testing phase in which the primary function is to build a prototype, test it to make sure it performs how we need it to, and certify it before starting our manufacturing phase. After certification is awarded and manufacturing logistics are worked out, we will enter our manufacturing phase. The first phase of manufacturing will set a goal of producing 50 vehicles per year for six years with one production line in South Carolina, with the first vehicle ready for service in 2037, before ramping up production to 100 planes for seven years with two production lines in South Carolina spanning into 2049 to total 1000 units. A visual plan of this timeline can be seen below in Figure 3.47.

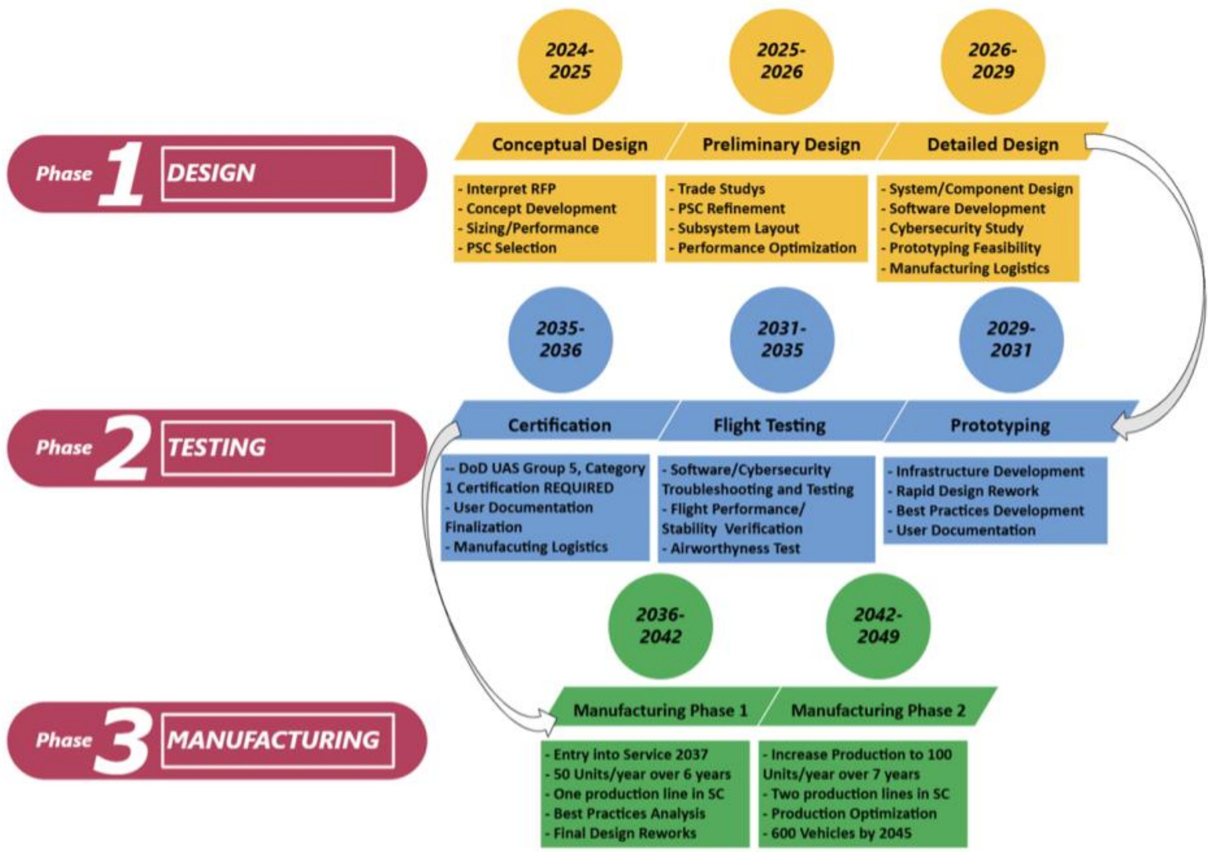


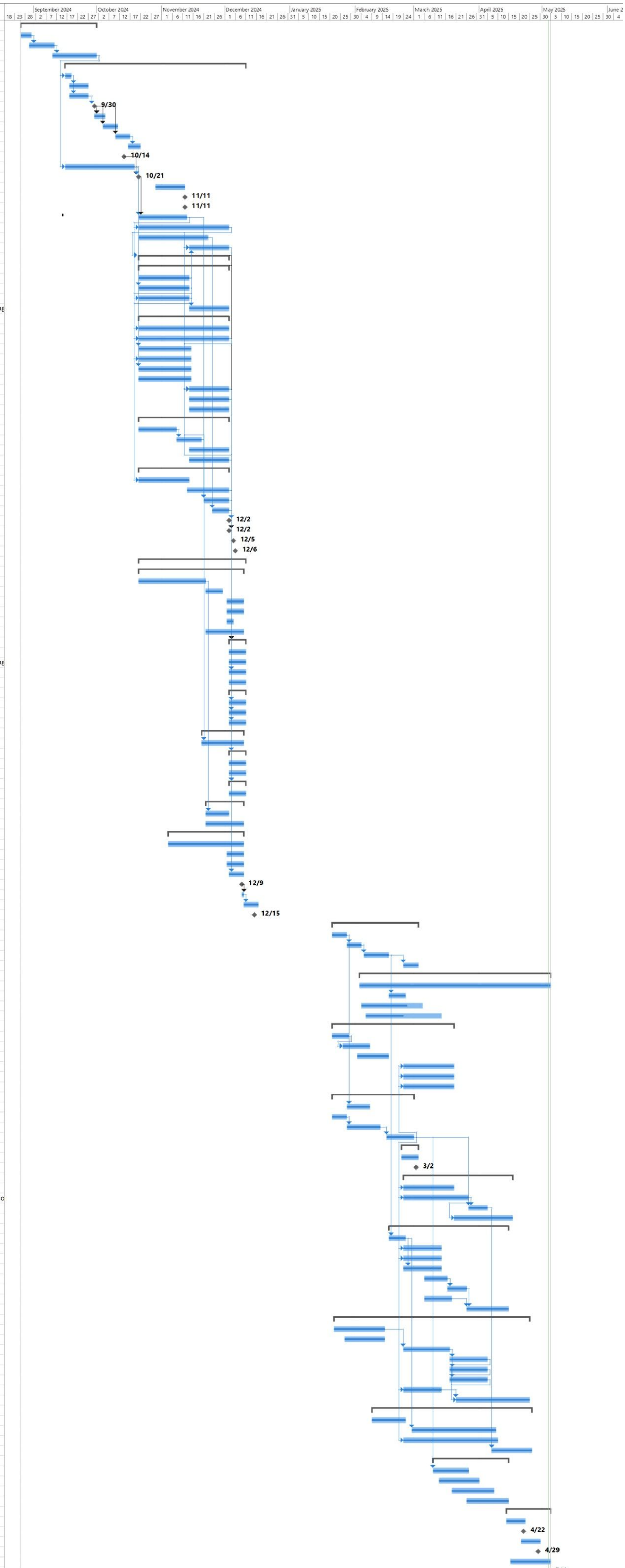
Figure 3.47: EIS Timeline

This timeline considers many factors that may hinder our efficiency going forward. For instance, we have allotted ourselves ample time during the conceptual and preliminary design phase to center our design down to exactly what we want before we start designing the individual parts and subsystems through the detailed design phase. Cybersecurity is also a factor we have heavily accounted for, as it will be one of our greatest threats for this vehicle. To mitigate this, we have significant software development starting in our detailed design phase, spanning throughout the testing phase with penetration testing as a key focus as we work towards certification. Lastly, this production plan is optimized to reflect our stringent cost constraints, as the costs will rapidly lower as we break the 25 million dollar cost per plane by the 339th unit produced. As such, once sufficient best practices are studied for the production of this vehicle, a second production line will be utilized, modeling the first production line for the remainder of the manufacturing phase.

3.14.5 Gantt Chart

To maintain organization and track progress, the team utilized a Gantt chart created in Microsoft Project. The Gantt chart splits up the project into smaller more manageable tasks and helps visualize the interdependencies between tasks. Start dates, end dates, percent completion, predecessors, and owners can be specified for each task. Project milestones such as the PDR presentation are designated with a dark grey diamond whereas tasks are designated as blue bars. A complete Gantt chart for the work during this project can be found on the following page.

ID	Task Name
1	**Organization Setup**
2	Team Formation
3	Team Contract
4	Team Responsibilities/Structure
5	**Conceptual Design**
6	IOC & Comparator
7	MoMs
8	HoQ
9	SRR Review
10	EIS Timeline
11	WBS
12	N2 Diagram
13	Gannt Chart
14	MPP Review
15	Geometry/Sketches
16	SCR Presentation
17	Design Concepts Finalized
18	PRM
19	Mentor Meeting #2
20	Concepts Weight and Constraint Analysis
21	Concepts Sizing and Mission Performance Analysis
22	AOE-4105 Experiments Research
23	Concepts fully modeled in OpenVSP
24	**Design Characteristics to populate HOQ**
25	**Concepts Utilities and Subsystem**
26	Landing Gear location
27	Weapons Location
28	Avionics Location
29	Nose/main gear balance, tipover angle, and rollover ang
30	**Concepts Aerodynamics and Performance**
31	Wing Loading
32	Aerodynamic Sizing
33	Wings Geometry
34	Airfoil Selection
35	Tail Geometry
36	Prelim Control Surfaces Sizing
37	General Flight Performance
38	Drag Polar
39	Static Margin
40	**Concepts Propulsion**
41	Propulsion R&D
42	Engine Selection
43	Fuel Tank Location
44	Prelim Intake Design
45	**Concepts Mass Properties**
46	Component Weight Estimations
47	Weight Distribution/CG Location
48	Concept Cost Estimation
49	AOE-4105 Experiments Final Presentation Powerpoint
50	AOE-4105 Experiments Final Presentation
51	PSC Chosen
52	PRM
53	Mentor Meeting #3
54	**CODR Components**
55	**SCR Feedback and Overlapping Components**
56	Cybersecurity Research
57	Org Chart Revision
58	Measures or Merit (MoM) Revision
59	Key Performance Requirements (KPR)
60	ConOps Revision
61	Comparator Aircraft Revision
62	**PSC Utilities and Subsystems**
63	Landing Gear Layout
64	Nose/main gear balance, tipover angle, and rollover ang
65	3D View of Subsystems location
66	Subsystems Architecture Diagram
67	**PSC Aerodynamics, Stability, and Performance**
68	Drag Polar
69	Scissor Plot
70	Thrust vs. Velocity
71	**PSC Propulsion**
72	Engine Deck
73	**PSC Structures**
74	Component Weight Breakdown
75	CG Travel
76	**PSC Cost Analysis**
77	Unit Cost
78	**Risk Matrices**
79	Cybersecurity Risk
80	Program Risk
81	**Trade Studies**
82	Oblique wing vs non-oblique
83	2D Trade Study
84	Payload-range diagram
85	Aircraft Walkaround
86	Conceptual Design Review
87	Analyze CoDR feedback
88	Draft Final Report
89	Final Report Due
90	**Mission Systems**
91	Resarch Ground Control System for OV Diagrams
92	Research Kill-Chain and Cyber Resources
93	Refine OV Diagrams/CONOPS
94	Day in the Life Diagram
95	**Performance**
96	Mathworks Requirements Integration
97	Refine Requirements from CONOPS
98	Modeling in FLOPS
99	Modeling in DATCOM
100	**Propulsion**
101	Propulsion System Integration Research
102	Low Fidelity CAD model of Engine
103	Size/Design Dual Inlet & Capture Area
104	Fuel Distribution
105	Fuel Pumps
106	APU Integration
107	**Utilities and Subsystems**
108	Research Avionics Suite
109	Research Landing Gear Modeling
110	Model High-Fidelity Landing Gear
111	Detailed Subsystems Layout
112	**iPDR Work**
113	Construct Presentation
114	iPDR Due Date
115	**Mass Properties**
116	Component Weights
117	Refined c.g. travel in all axes for all payload configuratic
118	Moment of Inertial Calculations
119	Potato Plot
120	**Aerodynamics**
121	Define aero requirements based on CONOPS
122	Revise OpenVSP model
123	Tail Sizing
124	Flaperon Sizing
125	Parachute Assembly Design
126	Parachute Performance Analysis
127	Inlet Analysis
128	Complete Aerodynamics Analysis
129	**Structures**
130	Research Wing Box Design
131	V-N Diagram
132	Wing Box Design and Analysis
133	Fuselage Structural Design
134	Wing Structural Design
135	Control Surface/Stabilizer Structural Design
136	Component Housings
137	Complete Load Anlysis/FEA
138	**Stability and Control**
139	Control Modeling in Matlab Research
140	Empanage and Control Surface Sizing
141	Static Stability Analysis and Refinement
142	Dynamic Stability Analysis and Refinement
143	**Cost Analysis**
144	Component Cost Breakdown
145	Refined Unit Cost
146	Operating Costs
147	Manufacturing Cost Estimations
148	**PDR Work**
149	Construct Presentation
150	PDR Due Date
151	Make Expo Poster
152	Expo Poster Due
153	Write Final Report
154	Final Report Due Date



References

[1] ASM, "ASM material Aluminium 2024-T6 data sheet," .

[2] ASM, "ASM material Aluminum 7075-T6 data sheet," .

[3] Nicolai, L. M., Carichner, G., and Nicolai, L. M., *Fundamentals of aircraft and airship design*, AIAA educational series, American Institute of Aeronautics and Astronautics.

[4] United States Air Force, "F-16 Fighting Falcon," .

[5] BAE Systems, "What is JADO?" .

[6] BAE Systems, "What does JADC2 stand for?" .

[7] DAF PEO C3BM Public Affairs, "Advanced Battle Management System: victory through distributed connectivity," .

[8] Leone, D., "Remembering the F-16 fighter jet modified to test the F-35's diverterless supersonic inlet," .

[9] Laurence K. Loftin, J., *Theoretical and Experimental Data for a Number of NACA 6A-SEPIES Airfoil Sections*, National Advisory Committee for Aeronautics.

[10] Adam Entsminger, David Gallagher, Will Graf, "General Dynamics F-16 Fighting Falcon," .

[11] HAW-Hamburg, "Drag Prediction," .

[12] Brandt, S. A., editor, *Introduction to aeronautics: a design perspective*, AIAA education series, American Institute of Aeronautics and Astronautics, 2nd ed.

[13] Raymer, D. P., *Aircraft design: a conceptual approach*, AIAA education series, AIAA, American Inst. of Aeronautics and Astronautics, 5th ed.

[14] AIAA, "2024-2025_team_homeland_defense_interceptor-updated-9-december-2024," .

[15] Sadraey, M. H., *Aircraft design: a systems engineering approach*, Aerospace series, Wiley.

[16] Moorhouse, D. J. and Woodcock, R. J., "Background Information and User Guide for MIL-F-8785C, Military Specification - Flying Qualities of Piloted Airplanes," .

[17] Air Force, "F-15 Eagle," .

[18] GE Aviation, "F110-GE-129 Spec Sheet," .

[19] Eurofighter, "Features — eurofighter typhoon," .

[20] NAVAIR, "F/A-18 A-D hornet: Navair," .

[21] Mattingly, J. D., *Elements of gas turbine propulsion*, McGraw-Hill Education (India), 7th ed., OCLC: 914523412.

[22] Roskam, J., *Airplane design. 8: Airplane cost estimation: design, development, manufacturing and operating*, DARcorporation, 3rd ed., Num Pages: 368.